

Questions Before Priors: Causal Dissonance, Insight, and the Origin of Strategic Theories*

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— First draft —

Abstract

The theory-based view treats economic actors as theorists whose causal understandings direct search, experimentation, and action. Existing formal treatments begin after the theory exists. This paper models where strategic theories come from. Agents represent their environment through awareness partitions of a state space; expressible causal theories assign elementary mechanisms to awareness cells; theories and beliefs determine standing policies; and the policy profile causes everyone's consequences through an objective causal system. Bayesian learning operates only within the current language, and a misspecified posterior grows confident rather than alarmed, so detecting representational failure requires a distinct faculty: causal dissonance, the statistical rejection of every expressible theory. Dissonance directs inquiry without determining its output; a costly, fallible technology may produce a previously inexpressible distinction; and a bridge prior restores Bayesian competence on the expanded language. An agent can estimate the value of a question before its answers are conceivable, warrant is distinct from post-insight confidence—the insight itself is not evidence—and an agent generally cannot measure its own overconfidence. Because theories cause actions and actions generate evidence, inquiry dies out almost surely, leaving every agent either in a Bayesian world or in rationally tolerated dissonance—its question unpriced, unanswerable, or abandoned. The configurations that remain can preserve false theories, heterogeneous awareness, and permanent performance differences: shared false theories survive exactly when neither strategic variety nor environmental harshness exposes them, and performance gaps persist whenever the laggard's question is priced below the cost of asking.

Keywords: theory-based view; unawareness; insight; causal learning; subjective rationality; self-confirming equilibrium

*Preliminary draft; comments welcome. This paper develops the formal core of a research program on pre-Bayesian theory formation in strategy.

1 Introduction

The theory-based view of strategy holds that economic actors are theorists: managers and entrepreneurs carry causal accounts of how value is created, and those accounts—not merely their information or their incentives—direct what they notice, which experiments they run, and what they do (Felin and Zenger, 2017; Felin et al., 2024). A decade of work has established the consequences of holding a theory. Theories guide experimentation and search (Camuffo et al., 2020; Chavda et al., 2024), support the acquisition of resources and rents (Ehrig and Zenger, 2024), and discipline entrepreneurial decision-making in the field (Camuffo et al., 2024). What this literature has not yet formalized is the step before all of that: where a genuinely new theory comes from. Existing formal treatments begin with the theory present—as a menu of candidate theories to select among (Camuffo et al., 2024), as a given causal conjecture that directs search (Chavda et al., 2024), or as a given awareness advantage to be exploited competitively (Ehrig and Zenger, 2024). The origin of the theory is described verbally and documented empirically (Ott and Hannah, 2024), and the absence of formal machinery for it has been explicitly noted as the field’s open analytical problem (Felin et al., 2024).

The difficulty is not a missing application of Bayesian reasoning; it is a boundary of Bayesian reasoning. A Bayesian agent updates beliefs over a space of represented possibilities. Conditioning reweights that space and cannot enlarge it, so no sequence of updates makes an unrepresented causal distinction available for belief or action. Worse, when every represented theory is false, the posterior does not sound an alarm: by the logic of misspecified Bayesian learning (Berk, 1966), it concentrates on the least wrong of the available theories, and the agent grows more confident, not less. Within-language updating is self-sealing. The economics of growing awareness has developed axiomatic accounts of how beliefs should extend when the state space expands (Karni and Vierø, 2013, 2017; Vierø, 2021), but in those models the expansion itself arrives exogenously; models of discovery in games describe how play can reveal objectively available actions (Schipper, 2021) without explaining how experience generates a substantive causal question or its answer. What is missing, in both the strategy literature and the economics of unawareness, is a model of the transition: from a broken causal language, through a directed question, to a new representation, and back to ordinary Bayesian life.

This paper models that transition, embedded in strategic interaction. Its organizing object is the chain

$$\begin{array}{ccccccc} \text{theory} & \longrightarrow & \text{Bayesian judgment} & \longrightarrow & \text{action} & \longrightarrow & \text{experience} & \longrightarrow & \text{dissonance} & \longrightarrow \\ & & \text{inquiry} & \longrightarrow & \text{refined awareness} & \longrightarrow & \text{new models} & \longrightarrow & \text{theory} & \longrightarrow \dots \end{array}$$

the process by which an agent whose causal representation has failed regains the capacity to be a Bayesian. The back end of the model follows the subjective-rationality tradition (Kalai and Lehrer, 1993b; Ryall, 2003): a finite set of agents adopt causal theories, combine them with beliefs

and conjectures to choose standing policies, and the policy profile determines every agent's consequences through an objective causal system that absorbs all market and environmental detail. The front end is new. Agents represent the environment through awareness partitions of a state space; the theories they can formulate assign elementary causal mechanisms to awareness cells; and the partition itself—the language—is endogenous. An awareness cell is an implicit claim that the states it pools are causally alike. Strategic experience can refute that claim: when the statistical record rejects every expressible theory, the agent faces *causal dissonance*, a determinate defect with no expressible repair. Dissonance makes a question salient without providing its answer. A costly, fallible, truth-blind inquiry technology may then produce a refinement of the partition—the formal footprint of an insight—after which a bridge prior restores a probability space and Bayesian judgment resumes on the expanded language.

Five sets of results give the chain its content. First, boundary results locate exactly where Bayesian competence stops: conditioning cannot expand the language, and the misspecified posterior is self-sealing, so detecting representational failure requires a faculty formally distinct from updating. Second, dissonance directs inquiry without determining it. The record restricts admissible repairs—every restorative refinement must separate the evidence clusters that no single mechanism can accommodate—but generally many minimal repairs remain, observationally equivalent worlds demand different answers, and no truth-blind technology can be uniformly correct. Fallibility is the price of a representation that economizes on distinctions. Third, an agent can price a question before its answers are conceivable. A leverage statistic built from the agent's own retained experience—its remembered tokens and the coherent groupings it can articulate, never the analyst's list of repairs—estimates the value of inquiry at exactly the moment the acting belief is undefined, and it tracks whether an answer could change action, not how much uncertainty remains. The estimate is honest: under noise the agent cannot tell a causal contrast from sampling accident, and the achievability weight it applies decays as inquiry fails. Questions die rationally: sustained success extinguishes them, defeat concentrates the evidence and the incentive in the loser, and hard questions are eventually abandoned. Fourth, the return to Bayes is disciplined by a warrant account: because any successful repair fits the experience that generated it by construction, retrospective fit carries no information, and—when inquiry is truth-blind—neither does the arrival of the insight itself. Post-insight confidence therefore exceeds warrant by a gap of size an outside assessor can compute, and an agent holding a minimal repair generally cannot even formulate the comparison that would measure its own gap. Fifth, because theories cause actions and actions generate the evidence, the population dynamics close the loop. Almost surely, awareness change and inquiry die out, leaving every agent either inhabiting a Bayesian world—models fit, questions dead—or in rationally tolerated dissonance, its question unpriced, unanswerable, or abandoned. We call the fully-Bayesian configurations *self-confirming unawareness equilibria*, and their interest is epistemic rather than strategic: such a configuration can preserve false theories and heterogeneous awareness, and

a shared false theory survives precisely when neither of the two corrective forces—strategic variety inside an agent’s categories, or an environment harsh enough to punish misservice observably—is present.

The persistence results deserve emphasis because of what carries them. Performance differences persist in this model not because an equilibrium concept freezes them but because the process that would erase them has died economically: the disadvantaged agent’s question generates no leverage, loses its achievability weight, or meets an irreparable record—in each case it is priced below the cost of asking. Strategic variety is therefore traced to its source. Agents with identical preferences, identical data-processing competence, and identical objective opportunities end up with different causal languages, different theories, and different payoffs because their experiences posed different questions, their technologies produced different repairs, and their subsequent positions extinguished further inquiry at different points. This is a microfoundation for heterogeneity that begins before probability does.

A deliberately small example runs through the paper—two firms, four customer states, two product architectures—in which every object can be computed exactly: the four minimal repairs of a broken language, the two that survive a simplicity filter, the impossibility of choosing between them from the data, the exact price of the question, the exact warrant gap, and the knife-edge condition under which a market’s shared false theory protects itself forever. The example is a laboratory, not the theory; each of its results instantiates a general statement.

The paper proceeds as follows. [Section 2](#) locates the contribution. [Section 3](#) presents the model. [Section 4](#) develops the Bayesian layer and its boundary; [section 5](#) defines dissonance and exposure; [section 6](#) analyzes restoration, direction, and fallibility; [section 7](#) prices inquiry; [section 8](#) develops formulation, warrant, and the return to Bayesian competence; and [section 9](#) closes the loop with the population dynamics. [Section 10](#) discusses implications and the research program.

2 Related literature

The theory-based view. The foundational claim is that theories do more than summarize evidence: they direct attention toward causal possibilities, support counterfactual reasoning, and guide experimentation ([Felin and Zenger, 2017](#)). The *Strategy Science* collection on theory-based decisions substantially developed the view ([Felin et al., 2024](#)): [Camuffo et al. \(2024\)](#) formalize the choice of which theory to test and show that the value of experimentation grows with the number and uncertainty of candidate theories; [Chavda et al. \(2024\)](#) model search guided by a given causal theory; [Ehrig and Zenger \(2024\)](#) translate theories into awareness structures and show how partial disclosure secures resources and rents; [Sorenson \(2024\)](#) and [Adner and Levinthal \(2024\)](#) probe the mechanisms and the limits of theory-guided experimentation; and [Ott and Hannah \(2024\)](#) document empirically how entrepreneurs develop a poorly formed thesis

into an articulated causal model. A parallel empirical program shows that training entrepreneurs to formulate and test theories improves performance (Camuffo et al., 2020). Related conceptual work presses the same open question this paper addresses: Zellweger and Zenger (2023) treat entrepreneurs as scientists generating beliefs under uncertainty, and Felin and Holweg (2024) argue that forward-looking causal theorizing is precisely what data-driven prediction cannot supply. Throughout, the formal treatments condition on the theory, the attribute list, or the awareness advantage being present. This paper supplies the front end those models presuppose, and its equilibrium analysis shows when the supplied front end matters: heterogeneous causal languages, and the performance differences they support, persist exactly when the strategic and environmental conditions identified below fail to erase them.

Unawareness and growing awareness. Standard state-space models cannot represent nontrivial unawareness (Dekel et al., 1998); generalized structures represent multiple awareness levels and interactive beliefs (Heifetz et al., 2006, 2013). On belief revision under expansion, reverse Bayesianism preserves relative odds as the state space grows (Karni and Vierø, 2013), with extensions to awareness of unawareness (Karni and Vierø, 2017), intertemporal settings (Vierø, 2021), unforeseen evidence (Piermont, 2021), and learning under incomplete awareness (Grant et al., 2022). Schipper (2021) distinguishes learning, which discards conceived possibilities, from discovery, which adds unconceived ones, and shows how play can move players to richer games; Schipper (2025) shows that beliefs about novelty can be held without representing the novel objects—a precedent for the delabeled constructs used here. Relative to this literature the paper endogenizes what those models take as given: the expansion event itself. Experience generates the defect, the defect localizes the question, a fallible technology produces the new distinction, and the bridge prior problem—which reverse Bayesianism constrains but does not solve—is embedded in an explicit account of what the generating evidence can and cannot warrant. In strategy, Bryan et al. (2022) bring unawareness into value capture and show that awareness itself earns rents; the present paper models where the awareness advantage originates, which that paper leaves open.

Causal learning with a fixed vocabulary. Causal graphical models separate intervention from association (Pearl, 2009; Spirtes et al., 2000); Ryall (2009) brings this machinery to strategy, modeling imitation as Bayesian inference over operating structures in which passive observation narrows candidates to an observational-equivalence class and experimentation may discriminate among survivors. Active-learning methods select interventions to split a represented equivalence class (He and Geng, 2008). These models are the benchmark for the paper’s second learning tier: inference over structures expressible in a fixed vocabulary. The paper begins where they stop—when no expressible structure is adequate—and its nonidentification results extend the equivalence-class logic to the choice among vocabulary repairs: observationally equivalent

worlds can demand different distinctions, so no truth-blind repair rule is uniformly correct.

Subjective rationality and self-confirming equilibrium. The back end follows Kalai and Lehrer (1993b) and Fudenberg and Levine (1993): beliefs correct on the path of play need not be correct off it, and subjectively optimal behavior against such beliefs can persist. Ryall (2003) showed that competitive advantage can rest entirely on rivals' self-confirming false theories. Those models fix the hypothesis space; convergence results in the rational-learning tradition require a grain of truth—the truth absolutely continuous with respect to beliefs (Kalai and Lehrer, 1993a). Under unawareness the grain-of-truth condition can fail at the level of language: the truth may be inexpressible. This paper supplies the missing dynamics. Statistical dissonance is the detector of language-level failure, awareness refinement restores the grain of truth on the path of play, and the fixed-language models are recovered exactly as the special case in which awareness never changes. The paper's absorbing configurations extend self-confirming equilibrium with two epistemic conditions—the language survives its own evidence, and no question is worth its price—and it is these conditions, not the strategic ones, that carry the new results.

Statement of contribution. Existing models show how theories guide learning, how Bayesian agents discriminate among represented causal structures, and how play can reveal previously unavailable actions. This paper models a different transition. An agent's causal language pools objectively different situations; strategic experience endogenously reveals an action-relevant failure of that pooling and thereby creates a localized open question; costly inquiry activates a fallible technology whose output is not measurable in the agent's prior awareness; only after that output is formulated does the agent construct a prior and undertake Bayesian judgment, under an explicit account of how much confidence the generating experience licenses. Because actions depend on theories, play can either create the contrasts that generate questions or suppress them—so the model also yields population-level sufficient conditions: for when shared false theories persist, and for when a diagnostic environment erases them.

3 The model

3.1 Agents, states, actions, and consequences

There are $n \geq 2$ agents, $N = \{1, \dots, n\}$. Time is a sequence of periods $t = 1, 2, \dots$. In each period one state is drawn, every agent acts through a *standing policy* carried over from the previous period, and consequences are realized. Agents update beliefs every period. If evidence mounts that their model of the world is in conflict with reality, they may engage in a process of costly inquiry which may then lead to refined awareness, changes to their held model, and potential revisions of their business strategy.

In each period a *state* $s \in S$, with S finite, is drawn independently from a full-support distribution $\rho \in \Delta(S)$. States describe the strategically-relevant features of the environment that evolve or change over time: customer configurations, input conditions, competitive circumstances, available production technologies, and so on. (Their status in the agents' own representations is the subject of [section 3.3](#).)

Contingent upon agent i 's awareness of the consequence-relevant features of his present state, he chooses an *action* a_i from a finite set A_i through the standing policy described below. An *action profile* is a list of each agent's action choice, $a = (a_1, \dots, a_n)$ where $A = \prod_i A_i$ is the set of all action profiles. Write $A_{-i} = \prod_{j \neq i} A_j$.

Finally, $X = \prod_i X_i$ is the finite set of performance-related *consequences* with typical element $x = (x_1, \dots, x_n)$. Agent i 's consequence x_i is everything a period delivers to it—profit, sales, rival behavior, and any other observed events. Then, agent i 's *payoff function* $\pi_i : X_i \rightarrow \mathbb{R}$ maps his consequences to payoffs.

3.2 The objective causal system

Causality is modeled in a general, high-level fashion. Each period, the state s , together with the agents' actions a , determines the consequences x .

Definition 3.1 (Elementary mechanisms and the objective system). *A finite set Θ of elementary causal mechanisms is given, each element of which is a map $\theta : A \rightarrow \Delta(X)$.¹ The objective assignment is a function $\tau : S \rightarrow \Theta$. Both Θ and τ are primitives of the model. The objective causal system is $\tau(s)(a) \in \Delta(X)$, a distribution over consequences determined by the state and action profile associated with a period. Two states s and s' are causally equivalent if $\tau(s) = \tau(s')$. Let $\mathcal{P}^\tau$ denote the partition of S into causal-equivalence classes. Finally, $\text{marg}_i \theta(a)$ denotes the X_i -marginal of $\theta(a)$.*

Thus, τ implies a distribution of consequences for every action profile at every state, including profiles never played and states never occurring. Agents' actions, together with the state, cause their consequences. A mechanism is the reduced form of whatever detailed model—a structural causal model of customer behavior, a market microstructure, a production process—underlies that relation; the paper imposes no structural-causal-model formalism, but examples may exhibit micro-foundations explicitly, as the running example's customer market does.

Three consequences of the causal reading recur. First, theories, defined next, are interventional objects: they make predictions at unplayed profiles. Second, on-path data constrain $\tau(s)(a)$ only at visited state–profile pairs, so observationally equivalent systems with different counterfactual content are generic; that is the engine of the identification results. Third, the on-path distribution of consequences given an agent's category of states is a *selection* object—it mixes mechanisms according to which states co-occur with which play—and must never be conflated with any mechanism's own conditional.

¹Here $\Delta(X)$ denotes the set of probability distributions on X . Deterministic causal effects are a special case.

3.3 Awareness, theories, and conjectures

Definition 3.2 (Awareness and expressible theories). *Agent i 's awareness partition is $\mathcal{P}_i \in \Pi(S)$, the set of partitions of S . The agent's represented causal and decision objects are restricted by $\mathcal{P}_i$: theories and policies are defined on its cells, conjectures condition only on its cells, and beliefs range only over theories expressible on it. The expressible theories are the cell-constant mechanism assignments*

$$M(\mathcal{P}_i) = \{m : \mathcal{P}_i \rightarrow \Theta\}, \quad (1)$$

with m predicting $x_i \sim \text{marg}_i m(C)(a)$ at cell C under profile a . Partition $\mathcal{P}$ is sufficient for an assignment τ' if τ' is constant on every cell of $\mathcal{P}$, equivalently if $\mathcal{P}$ refines $\mathcal{P}^{\tau'}$; the objective assignment is expressible for agent i if and only if $\mathcal{P}_i$ is sufficient for τ .

An awareness cell is thus an implicit causal claim: the states it pools are alike in how actions produce consequences. An expressible theory makes the claim explicit by naming the mechanism. Restricting theories to *single* mechanism assignments—rather than arbitrary conditional distributions per cell—is substantive and deliberate. If any distribution were expressible, any within-cell mixture would be too, and no experience could reject the class: coarseness alone cannot produce a detectable failure in a saturated model space. The mechanism vocabulary Θ is known to all agents from the outset; what an agent may lack is the *distinction* that separates states carrying different mechanisms. Everything the paper calls insight is the production of such a distinction. Expansion of the mechanism vocabulary itself is a deeper dimension of awareness growth deliberately left outside this paper.

Agent i holds a belief $\mu_i \in \Delta(M(\mathcal{P}_i))$ over $M(\mathcal{P}_i)$ —formally, the acting tier of a two-tier belief specified in [section 5](#)—and a conjecture $q_i : \mathcal{P}_i \rightarrow \Delta(A_{-i})$ about rival play that conditions on $\mathcal{P}_i$ -cells. Theories and conjectures are different kinds of objects and are deliberately separated: a theory is a causal claim about the objective system; a conjecture is a strategic claim about what other agents will do. The diagnostic apparatus developed below tests theories, not conjectures.

Assumption 3.3 (Observation, retention, and the two records). *At the start of each period agent i perceives its current awareness cell $C_{\mathcal{P}_i}(s)$, the cell of $\mathcal{P}_i$ containing s ; at the end it additionally observes the realized action profile a and its own consequence x_i . Each period is retained as a token. Two records are distinguished throughout. The deliberative record r_i is the multiset of tokens $(C_{\mathcal{P}_i}(s), a, x_i)$; it is the only record on which the agent's beliefs, tests, and decisions may depend. The analyst record $\bar{r}_i$ additionally carries the underlying state s of each token. No object used in the agent's deliberation may depend on the underlying state labels carried only by $\bar{r}_i$. Analyst-level constructions that require those labels—in particular the restorative correspondence and the warrant accounting of [section 8](#)—are so marked. The non-deliberative representation-producing technology is specified separately in [definition 6.4](#).*

Recodability. *When the agent acquires a new distinction—a refinement $\mathcal{P}'$ of its partition—it also acquires the capacity to classify each retained token under $\mathcal{P}'$, producing a recoded deliberative record. Before acquisition neither belief nor action can condition on that distinction. This capacity is a modeling*

idealization: it treats the retained particular as re-describable under any predicate the agent comes to possess, while denying the agent any pre-insight access to the predicate itself.

Retention without recodability-in-advance is the minimal separation between experiencing a particular and possessing a projectible concept for it: the agent can remember “customers like that one” without commanding a predicate that distinguishes them, and the technology of [section 6](#) operates on exactly this material. The idealization has a cost worth naming: recoding is treated as total and infallible once the predicate is held, whereas real reclassification of remembered particulars is partial and error-prone; [section 8](#) isolates where this matters. Observability of the full action profile keeps the separation of causal from strategic error clean; a variant folding components of a into x_i changes bookkeeping only. The start-of-period perception of the current cell is what lets a cell-contingent policy be implemented; the state s itself is never perceived.

Assumption 3.4 (Standing policy, subjective optimality, and costly switching). *Agent i carries a standing policy $\sigma_i : \mathcal{P}_i \rightarrow A_i$ from one period to the next. Given its current acting belief μ_i and conjecture q_i , the belief-optimal policy σ_i^* plays at each cell C*

$$\sigma_i^*(C) \in \arg \max_{a_i \in A_i} \sum_{m \in M(\mathcal{P}_i)} \mu_i(m) \sum_{a_{-i} \in A_{-i}} q_i(a_{-i} \mid C) \mathbb{E}_{x \sim m(C)(a_i, a_{-i})} [\pi_i(x_i)], \quad (2)$$

For every cell C of its current awareness partition, agent i knows the induced cell probability $\rho(C)$. When a refinement is acquired, the probabilities of the newly representable cells become available with that refinement. This does not give the agent access to the decomposition of a current cell into states or distinctions it does not yet represent. The associated subjective per-period value is

$$U_i(\sigma \mid \mu_i, q_i) = \sum_{C \in \mathcal{P}_i} \rho(C) \sum_{m \in M(\mathcal{P}_i)} \mu_i(m) \sum_{a_{-i} \in A_{-i}} q_i(a_{-i} \mid C) \mathbb{E}_{x \sim m(C)(\sigma(C), a_{-i})} [\pi_i(x_i)], \quad (3)$$

where $\rho(C) = \sum_{s \in C} \rho(s)$. Agent i values a lumpy, one-time cost against the gain it produces over a fixed valuation horizon of T periods, common across agents, without discounting—the planning horizon over which a one-time cost is amortized, a bounded-rationality stand-in for discounting. Revising the standing policy is costly: agent i replaces σ_i by σ_i^ if and only if the horizon- T gain clears a one-time switching hurdle $\kappa_i \geq 0$,*

$$T(U_i(\sigma_i^* \mid \mu_i, q_i) - U_i(\sigma_i \mid \mu_i, q_i)) > \kappa_i,$$

and otherwise carries σ_i unchanged. The inaction band $T[U_i(\sigma_i^) - U_i(\sigma_i)] \leq \kappa_i$ is the source of policy inertia: beliefs may drift and the belief-optimal action may move, yet the standing policy holds until the gain from switching clears the hurdle. When dissonance ([section 5](#)) leaves μ_i undefined, σ_i^* is undefined and the standing policy is carried forward unchanged.*

Policy inertia is thus an outcome, not an assumption: a standing policy persists until the belief-optimal action moves far enough to clear the switching hurdle κ_i , or until the language

itself breaks down and must be repaired (section 5). How long play stays fixed is therefore governed by agent-level parameters—the switching hurdle here, and the tolerance for misfit introduced with the test of section 5—rather than fixed by assumption.

3.4 On-path conditionals

Fix a stable stretch over which the standing profile $\sigma = (\sigma_1, \dots, \sigma_n)$ is constant. Each state s receives the deterministic profile $a(s) = (\sigma_j(C_{\mathcal{P}_j}(s)))_{j \in N}$. Agent i 's *on-path conditional* at an observation pair (C, a) with $S(C, a) = \{s \in C : a(s) = a\} \neq \emptyset$ is

$$p_i^*(\cdot \mid C, a) = \sum_{s \in S(C, a)} \rho(s \mid S(C, a)) \text{ marg}_i \tau(s)(a). \quad (4)$$

This is the selection object flagged above: rivals with finer partitions condition on distinctions agent i cannot see, so the within-cell state mixture can vary with the profile. It is precisely this dependence—differentiated play by others inside one's own categories—that generates the contrasts on which the diagnostic apparatus feeds.

3.5 Modeling commitments

Five disciplines govern everything that follows; they are what distinguish a model of unawareness from a model of inattention or model selection. (i) Agents hold no prior over the outputs of the inquiry technology and never optimize over named future theories; pre-insight deliberation quantifies only over represented objects. (ii) Questions are generated by defects in the current representation, not drawn from a primitive list. (iii) Inquiry can fail, and its successes can be false; nothing reveals the objective assignment by construction. (iv) All valuations attached to inquiry are subjective, coarse, and possibly wrong; no objective payoff information leaks from behind the awareness frontier. (v) Evidence that generates a candidate representation is separated, in the formal accounting, from evidence that warrants it.

3.6 The running example

Example 1 (A four-state laboratory). Two firms serve a unit market. States are customer configurations $S = \{s_{00}, s_{01}, s_{10}, s_{11}\}$, displayed as a 2×2 array with row coordinate r and column coordinate c ; ρ is uniform. Actions are product architectures $A_i = \{0, 1\}$. The mechanism vocabulary is $\Theta = \{\theta^0, \theta^1, \theta^n\}$, with a demand parameter $\beta \in [0, 1]$: under θ^a ("architecture a is preferred"), if the firms differentiate, the firm offering a receives payoff 1 and its rival 0; if they pool on a , each receives $\frac{1}{2}$; if they pool on the other architecture, each receives $\beta/2$, the customer withholding $1 - \beta$ of the unit value. Under θ^n ("architecture is neutral"), each firm receives $\frac{1}{2}$ regardless. A consequence x_i is the pair (own payoff, observed action profile). The objective assignment is the *row law* $\tau(s_{rc}) = \theta^r$: the row coordinate is causally relevant, the

column coordinate is not. Both firms begin with the coarse partition $\mathcal{P}_i^0 = \{S\}$, under which the expressible theories are the three constant assignments; the row law is not among them, because it is not constant on S . The baseline sets $\beta = 1$ (deterministic winner-take-all with market splitting); $\beta < 1$ returns in [section 9](#). All mechanisms here are deterministic, so every test below operates in its exact, zero-tolerance instance; nothing in the general development relies on that.

4 Bayesian competence and its boundary

Within fixed awareness, learning is ordinary: the Bayesian belief tier updates by Bayes' rule, while conjectures are updated empirically from observed play. [Section 5](#) distinguishes the reference posterior from the acting belief once test-based exclusions are introduced. If no agent's awareness ever changes, the model is a repeated interaction among subjectively rational Bayesian learners in the sense of [Kalai and Lehrer \(1993b\)](#) and [Ryall \(2003\)](#), and its stable configurations are self-confirming equilibria ([Fudenberg and Levine, 1993](#)). Everything new in this paper happens at the boundary of that layer, which two results locate.

Proposition 4.1 (No Bayesian emergence). *Any well-defined judged or ordinary conditioning of a prior $\mu \in \Delta(M(\mathcal{P}_i))$ on an observation in the agent's observation algebra yields a posterior $\mu' \in \Delta(M(\mathcal{P}_i))$. No sequence of such updates makes a theory outside $M(\mathcal{P}_i)$ expressible or a distinction finer than $\mathcal{P}_i$ usable in a policy.*

Proof. Conditioning reweights a distribution on its domain and cannot enlarge the domain; every element of $M(\mathcal{P}_i)$ is cell-constant, so every posterior mixture, and every policy optimal against one, is $\mathcal{P}_i$ -measurable. $\square$

Remark 4.2 (The posterior is self-sealing). *The stochastic setting yields a sharper point. Suppose every expressible theory is false: the agent's cell mixes mechanisms, so no cell-constant assignment matches the on-path conditionals. A full-support posterior does not signal this. By the logic of misspecified Bayesian learning ([Berk, 1966](#)), it concentrates on the expressible theories closest to the on-path conditionals in Kullback–Leibler divergence, and the agent becomes increasingly confident in a wrong theory. Posterior mass always sums to one; the inadequacy of the whole model class is not an event in the agent's algebra. Detecting that the language itself is broken therefore requires a faculty outside the posterior—the consistency test of the next section. The distinction matters for how the theory-based view should treat “updating”: ranking the answers one can express and judging whether any expressible answer is adequate are different cognitive operations, and only the first is Bayesian.*

Example 1 (continued). Under the coarse partition, firm i 's expressible theories are the constant assignments m^0, m^1, m^n . The row law τ is inexpressible: no reweighting of $\{m^0, m^1, m^n\}$, on any evidence, produces it ([proposition 4.1](#)). After successive deterministic observations

eliminate the expressible theories one by one, the observation that eliminates the last theory with positive posterior mass produces a zero Bayesian normalizer, so ordinary conditioning itself is undefined—the deterministic instance of [remark 4.2](#), in which misspecification announces itself only because likelihoods hit zero exactly. With stochastic full-support mechanisms, the Bayesian normalizer remains positive and no such announcement occurs.

5 Causal dissonance

5.1 Tests and active sets

Definition 5.1 (Sequential adequacy test, dissonance capacity, and active set). *Fix an adequacy tolerance $\eta \geq 0$. Theory m is adequate at a visited pair (C, a) when $\text{TV}(p_i^*(\cdot \mid C, a), \text{marg}_i m(C)(a)) \leq \eta$, where TV is total-variation distance; under a standing profile call m η -adequate if it is adequate at every visited pair and η -inadequate if it is inadequate at some visited pair. The agent monitors adequacy sequentially, period by period. For each theory admitted when the current language is initialized, it maintains a nonnegative evidence process E_i^m , initialized at 1 with that language; under adequacy, E_i^m is a nonnegative supermartingale, while at a recurrent pair where m is strictly inadequate the construction in [appendix A](#) contains a fixed betting component with strictly positive expected log-growth. Because the total evidence process assigns that component positive weight, E_i^m diverges almost surely under recurrent strict inadequacy. (The exact deterministic case $\eta = 0$ instead uses the separate zero-likelihood detector of [appendix A](#): a theory is rejected at the first realized consequence to which it assigns probability zero.) Agent i carries a dissonance capacity $b_i > 0$ and rejects m , permanently, at the first period in which $\log E_i^m \geq b_i$. The active set $\hat{M}_i(\mathcal{P}_i, r_i)$ contains the theories admitted when the current language was initialized that have not subsequently been rejected by the sequential test. On the initial language all theories in $M(\mathcal{P}_i)$ are admitted and each begins with $E_i^m = 1$; after a refinement, admission is determined by the initial repair screen described in the judged-updating paragraph below. Capacity is the agent’s evidence threshold for abandoning a theory: for a theory adequate from initialization, Ville’s inequality gives $\Pr(\sup_t E_i^m(t) \geq e^{b_i}) \leq e^{-b_i}$, so a high-capacity agent tolerates apparent misfit and rarely rejects on noise, while a low-capacity agent rejects on slight evidence and suffers more false alarms. Retrospective fits to a fixed record score a candidate against evidence already in hand rather than a stream, using the record’s counts $n_i(C, a)$ and empirical conditionals $\hat{p}_i(\cdot \mid C, a)$ and a nonnegative, nonincreasing slack $w(\cdot) \rightarrow 0$: theory m fits at a pair (C, a) when*

$$\text{TV}(\hat{p}_i(\cdot \mid C, a), \text{marg}_i m(C)(a)) \leq \eta + w(n_i(C, a)). \quad (5)$$

The repair test of [section 6](#) imposes this at every pair with $n_i(C, a) > 0$; the grouping test of [section 7](#) imposes it wherever $n_i(C, a) \geq \bar{n}$, a minimum count $\bar{n} \geq 1$.

Lemma 5.2 (Test consistency). *Fix a partition $\mathcal{P}_i$ and a standing profile, and suppose every theory is either η -adequate or strictly inadequate at each visited pair (none sits exactly at the adequacy boundary).*

Let t_0 denote the beginning of the stable standing-profile regime after a finite prefix, and let $E_i^m(t_0)$ be the inherited evidence value of a theory m active at t_0 . Because theories inactive at initialization never re-enter, sequential rejection is permanent, and the theories are finite, the active set is monotone and eventually constant; and almost surely:

- (a) every theory inactive at t_0 remains absent, and every theory active at t_0 that is strictly inadequate at a recurrent post- t_0 pair is eventually rejected and absent from the active set; and
- (b) a theory active at t_0 and η -adequate at every recurrent post- t_0 pair has conditional probability of future false rejection at most $E_i^m(t_0)e^{-b_i}$.

Active status implies $E_i^m(t_0) < e^{b_i}$, so every theory in (b) has strictly positive conditional probability of surviving forever. Consequently, if every expressible theory falls under (a), eventual dissonance occurs almost surely; if at least one recurrently adequate theory remains active at t_0 , eventual dissonance is not an almost-sure consequence; and if one such theory satisfies $E_i^m(t_0) \leq 1$, dissonance is avoided with probability at least $1 - e^{-b_i}$.

Proof. A recurrent pair is visited each period with a fixed positive probability under the i.i.d. full-support draw, hence infinitely often; a frozen pair is visited only in the finite prefix and affects the inherited value $E_i^m(t_0)$. If a theory m active at t_0 is strictly inadequate at a recurrent pair j^* , [appendix A](#) supplies a fixed betting component W^* for that pair with strictly positive expected log-growth. The pair mixture gives W^* positive weight π^* and the global evidence process gives the pair positive weight v_{j^*} , so pathwise $E_i^m(t) \geq v_{j^*}\pi^*W^*(t)$. The strong law gives $W^*(t) \rightarrow \infty$ almost surely, hence $E_i^m(t) \rightarrow \infty$ almost surely; therefore $\log E_i^m(t)$ eventually crosses b_i , and m is rejected. If m is active at t_0 and η -adequate at every recurrent post- t_0 pair, then $(E_i^m(t))_{t \geq t_0}$ is a nonnegative supermartingale and conditional Ville's inequality gives

$$\Pr\left(\sup_{t \geq t_0} E_i^m(t) \geq e^{b_i} \mid \mathcal{F}_{t_0}\right) \leq E_i^m(t_0)e^{-b_i}.$$

No re-entry, permanence of sequential rejection, and finiteness of $M(\mathcal{P}_i)$ make the active set monotone and eventually constant; collecting the two cases gives the dissonance dichotomy. $\square$

The inherited-value term matters: a theory adequate under the current profile may carry evidence from an earlier regime, so the active set and its future false-rejection bound track the whole retained process, not current play alone. The deterministic case uses the separate exact detector of [appendix A](#): when $\eta = 0$, an inadequate theory is rejected at the first diagnostic consequence to which it assigns probability zero, while an adequate theory is never falsely rejected. Capacity b_i plays no role in this detector and false alarms are zero.

Assumption 5.3 (Full-support marginals, stochastic case). *In the stochastic case every mechanism marginal $\text{marg}_i \theta(a)$ has full support on X_i . (The deterministic case, including the running example, is exempt and needs no such assumption.)*

The likelihood of a token (C, a, x_i) under theory m is $\text{marg}_i m(C)(a)(x_i)$: the agent’s subjective model treats the (C, a) -occurrence probability as theory-independent, so it cancels in Bayes’ rule and updating requires neither ρ nor the within-cell state mixture. That this occurrence probability is objectively theory-informative—rivals’ finer play correlates a_{-i} with the within-cell mechanism—is exactly the misspecification that feeds dissonance (remark 4.2); the agent does not model it.

Judged updating. Beliefs carry two tiers. The *reference posterior* $\bar{\mu}_i$ begins at a full-support prior on $M(\mathcal{P}_i)$ and updates by the token likelihoods above; it is never conditioned on the test. The *acting belief* μ_i is the reference posterior restricted and renormalized to the current active set, so that belief never rides on a rejected theory; when the active set is empty, the acting belief is undefined and the carry-forward clause of assumption 3.4 governs. Because the active set only shrinks after the current language is initialized, the acting belief is well defined at every period until—if ever—the last active theory is rejected; assumption 5.3 keeps every reference likelihood positive, so restriction to any nonempty active set is always to a positive-mass set. In the deterministic case judged updating coincides with ordinary Bayesian conditioning while the Bayesian normalizer is positive; a zero-likelihood contradiction can instead make ordinary conditioning itself undefined. After a refinement to $\mathcal{P}'_i$, the recoded record supplies an initial repair screen: the new-language active set is initialized as $\hat{M}_i^R(\mathcal{P}'_i, \bar{r}_i)$, which the agent can compute from its recoded deliberative record, and excluded theories begin inactive without replaying the historical record through the sequential evidence process. Each admitted theory begins with $E_i^m = 1$; the reference posterior begins at the full-support bridge prior $\mu_i^+ \in \Delta(M(\mathcal{P}'_i))$, and the acting belief is its restriction, renormalized on $\hat{M}_i^R(\mathcal{P}'_i, \bar{r}_i)$. Thereafter the reference posterior and sequential evidence processes update from newly arriving tokens, permanent sequential rejection further shrinks the active set, and the acting belief is the reference posterior restricted and renormalized to the current active set. Judged updating is the formal junction between the test and the acting belief—between judgment and understanding (remark 4.2)—and it guarantees, in section 9, that surviving acting beliefs are supported on adequate theories.

5.2 Dissonance and exposure

Definition 5.4 (Causal dissonance). *Agent i experiences causal dissonance at any period at which $\hat{M}_i(\mathcal{P}_i, r_i) = \emptyset$: no theory admitted on the current language remains unrejected. The expressible set does not disappear; every element is either excluded by the current language’s initialization screen or subsequently rejected by the sequential test. The immediate question has a determinate defect but not a determinate answer: what unrepresented difference among the pooled states explains the heterogeneity of their consequences?*

Dissonance is the model’s rendering of a question for intelligence. It is localized (the failing

cells are identified by the test), it is directed (an adequate repair must reconcile the record), and it is semantically open (no expressible object names the missing distinction). Whether dissonance ever occurs, however, depends on whether play makes the falsity of the language visible.

Definition 5.5 (Exposure and concealment). *Under a standing profile, awareness $\mathcal{P}_i$ is exposed if every $m \in M(\mathcal{P}_i)$ is η -inadequate. A standing profile is concealing for agent i if some expressible theory is η -adequate under it while false as a claim about τ at some state its cells cover.*

Proposition 5.6 (Exposure and dissonance). *Fix a standing profile and suppose no theory sits exactly at the adequacy boundary (lemma 5.2). If every theory that is η -adequate under the fixed profile remains active when that profile begins at t_0 , then $\mathcal{P}_i$ is exposed if and only if eventual dissonance occurs almost surely. If the profile is not exposed and m is an active η -adequate theory at entry, then, conditional on the entry history, dissonance is avoided forever with probability at least $1 - E_i^m(t_0)e^{-b_i}$; if some such m satisfies $E_i^m(t_0) \leq 1$, this bound is at least $1 - e^{-b_i}$.*

Proof. By lemma 5.2, every entry-active theory strictly inadequate at a recurrent pair is eventually rejected almost surely, while every theory inactive at t_0 remains absent. If all profile-adequate theories are active at t_0 , the eventual active set is therefore empty almost surely exactly when there is no η -adequate theory—that is, exactly when the profile is exposed. For a non-exposed profile, conditional Ville’s inequality bounds the future rejection probability of any entry-active adequate theory m by $E_i^m(t_0)e^{-b_i}$, so the stated survival and no-dissonance bounds follow. $\square$

The activity proviso is not cosmetic: the dynamics of section 9 generate records whose early periods were played under coarser partitions and different profiles, so a terminally adequate theory may already be absent because it was excluded when the current language was initialized or was subsequently rejected, while an active theory may inherit an evidence value above one.

Exposure has two sources, and the distinction organizes much of what follows. Heterogeneity inside a cell surfaces either because *other agents play differently across the pooled states*—strategic differentiation inside one’s categories, which changes the visited profiles and the induced mixtures—or because *the environment punishes uniform play across causally different states in an observable way*. Both reappear as the corrective forces in section 9.

Example 1 (continued). Let $\beta = 1$ and suppose the firms’ priors over the constant theories are heterogeneous: firm 1 considers θ^0 more likely than θ^1 , and firm 2 the reverse. A direct computation shows that, against any conjecture about the rival, offering architecture 0 is subjectively strictly dominant for firm 1 and architecture 1 for firm 2 whenever $\mu_i(m^a) > (2 - \beta) \mu_i(m^{a'})$; at $\beta = 1$ this is simple belief asymmetry. The firms therefore adopt differentiated constant policies. Take the valuation horizon $T = 2$ and condition on two periods that serve the diagonal states s_{00} and s_{11} under differentiated play—a positive-probability event under uniform ρ ; the appendix records how the leverage below scales with the realized composition. With differentiated actions, the winner-take-all payoffs reveal the objectively preferred architecture at each state: firm 1 wins

at s_{00} (payoffs 1, 0) and loses at s_{11} (payoffs 0, 1). No constant assignment is consistent with both observations— m^0 fails at s_{11} , m^1 at s_{00} , and m^n predicts one-half at both—so each firm’s active set empties: causal dissonance, in its exact deterministic form. Had the firms instead pooled on the same architecture at both states, each would have received one-half at every period, an outcome consistent with all three constant theories: the pooled profile is concealing, and coarse awareness would have survived its own evidence indefinitely. Belief heterogeneity, by inducing differentiated play, is what made the shared language’s failure visible.

6 Inquiry: restoration, direction, and fallibility

Dissonance poses a question the current language cannot answer. This section characterizes what a successful answer must accomplish, shows that the requirement directs inquiry without determining its output, and establishes that any inquiry process meeting the paper’s disciplines must be fallible.

6.1 Restorative refinements

For $\mathcal{P}, \mathcal{P}' \in \Pi(S)$, write $\mathcal{P} \prec \mathcal{P}'$ when $\mathcal{P}'$ strictly refines $\mathcal{P}$. The restorative correspondence is an *analyst-level* object: deciding whether a candidate refinement $\mathcal{P}'$ restores expressibility requires recoding tokens to $\mathcal{P}'$ -cells, which uses the analyst record $\bar{r}_i$ of [assumption 3.3](#). It is defined here for the analysis of what a successful answer must accomplish; [section 7](#) builds the agent’s own deliberative valuation separately.

Definition 6.1 (Restorative refinement). *Fix a candidate refinement $\mathcal{P}_i \prec \mathcal{P}'$ and recode the analyst record $\bar{r}_i$ to the deliberative record $r_i^{\mathcal{P}'}$ on $\mathcal{P}'$. Define the repair-consistent active set*

$$\hat{M}_i^R(\mathcal{P}', \bar{r}_i) = \{m' \in M(\mathcal{P}') : m' \text{ satisfies equation (6)}\},$$

where the repair test is

$$\text{TV}(\hat{p}_i(\cdot \mid C', a), \text{marg}_i m'(C')(a)) \leq \eta + w(n_i(C', a)) \quad \text{for every } (C', a) \text{ with } n_i(C', a) > 0. \quad (6)$$

Here the counts and empirical conditionals are computed from $r_i^{\mathcal{P}'}$. The refinement $\mathcal{P}'$ is restorative at $(\mathcal{P}_i, \bar{r}_i)$ if $\hat{M}_i^R(\mathcal{P}', \bar{r}_i) \neq \emptyset$. It is minimally restorative if no restorative $\mathcal{P}''$ satisfies $\mathcal{P}_i \prec \mathcal{P}'' \prec \mathcal{P}'$. Write $\mathcal{R}(\mathcal{P}_i, \bar{r}_i)$ for the set of minimally restorative refinements.

The repair test deliberately applies to every nonempty recoded cell–profile pair. The count floor $\bar{n}$ is not part of the repair test; it applies only to the deliberative grouping test of [section 7](#). Every nonempty recoded fragment remains subject to [equation \(6\)](#), while the larger finite-sample slack $w(n)$ at small n preserves the intended caution about sparse evidence. Refinement can

therefore reconcile observations but cannot delete their constraints. In the exact deterministic case $\eta = 0$ and $w \equiv 0$, the repair test is exact consistency at every observed pair.

Proposition 6.2 (Cellwise coherence). *For a candidate refinement $\mathcal{P}_i \prec \mathcal{P}'$, define*

$$\Theta_i^R(C' \mid \bar{r}_i) = \left\{ \theta \in \Theta : \begin{array}{l} \text{TV}(\hat{p}_i(\cdot \mid C', a), \text{marg}_i \theta(a)) \leq \eta + w(n_i(C', a)) \\ \text{for every } a \text{ with } n_i(C', a) > 0 \end{array} \right\}.$$

Call C' repair-coherent if $\Theta_i^R(C' \mid \bar{r}_i) \neq \emptyset$. Then $\mathcal{P}'$ is restorative if and only if every cell of $\mathcal{P}'$ is repair-coherent.

Proof. Because $M(\mathcal{P}') = \Theta^{\mathcal{P}'}$ and every constraint in [equation \(6\)](#) involves only the mechanism assigned to one cell,

$$\hat{M}_i^R(\mathcal{P}', \bar{r}_i) = \prod_{C' \in \mathcal{P}'} \Theta_i^R(C' \mid \bar{r}_i).$$

This product is nonempty exactly when every factor is nonempty. □

Coherence is the general form of a conflict structure. Token sets are *mutually incoherent* when no single mechanism accommodates their union; every restorative refinement must separate mutually incoherent sets, since a cell containing both would be incoherent. In the deterministic running example incoherence is a binary relation between labeled periods and restoration is a graph-coloring problem; in general, a mixture of three mechanisms may defeat every single mechanism even though each pair of tokens is individually accommodatable, so restoration is a hypergraph coloring. Two further structural facts matter. First, minimal restorations, when they exist, are typically multiple—as the example shows vividly. Second, existence itself is guaranteed only in the deterministic case.

Remark 6.3 (Irreparable records). *With deterministic mechanisms, every dissonant record is repairable. Under the exact calibration $\eta = 0$ and $w \equiv 0$, consider the singleton partition of S . At each state s and each observed profile a , every retained consequence was generated by the deterministic mechanism $\tau(s)$; hence $\tau(s)$ fits all observations recoded to $\{s\}$ exactly. The objective assignment therefore belongs to the repair-consistent active set on the singleton partition. If the current partition is already the singleton partition, dissonance is impossible; otherwise finiteness of $\Pi(S)$ ensures a minimally restorative refinement.*

With stochastic mechanisms this guarantee fails. Multiple observations at a single state may yield empirical conditionals for which no one mechanism satisfies [equation \(6\)](#) across the observed profiles. Because no refinement can split a state, such a record has $\mathcal{R}(\mathcal{P}_i, \bar{r}_i) = \emptyset$ and is irreparable: dissonance names a defect that no expressible refinement can repair. Irreparability is an analyst-level property: it is a fact about $\bar{r}_i$, and the agent, lacking the state labels, cannot in general tell an irreparable record from a repairable one—this is precisely the agent’s inability to distinguish causal heterogeneity from mechanism noise, developed in [section 7](#). When the record is irreparable the technology fails surely ([definition 6.4](#));

the resulting behavior is governed by the deliberative valuation and failure-discouragement of [section 7](#). If the deliberative record still admits a positive-leverage coherent grouping, the agent may elect a doomed inquiry until its achievability weight decays; if no coherent grouping prices the question, it declines immediately. Neither response requires the agent to recognize analyst-level irreparability. Under recurring visitation irreparability is typically transient, but a record whose critical pairs stop being visited can remain irreparable forever, a possibility [section 9](#) classifies within dissonant stasis.

6.2 The technology

Definition 6.4 (Representation-producing technology). A representation-producing technology for agent i is a possibly randomized map ϕ_i from the analyst record $(\mathcal{P}_i, \bar{r}_i)$ to $\{\mathcal{P}_i\} \cup \mathcal{R}(\mathcal{P}_i, \bar{r}_i)$: it fails, returning the current partition, or succeeds, producing a minimally restorative refinement. When $\mathcal{R}(\mathcal{P}_i, \bar{r}_i) = \emptyset$, the range is the singleton $\{\mathcal{P}_i\}$ and the technology fails surely. It is truth-blind: its output distribution depends only on $(\mathcal{P}_i, \bar{r}_i)$, never directly on τ . The technology consumes the analyst record because producing a distinction operates on retained particulars in their full concreteness; what the agent lacks pre-insight is not the particulars but the predicate that organizes them, and truth-blindness records that even this concrete operation cannot read off the objective law. A complexity ordering on refinements—the analyst’s model of which conceptual products the agent’s comparison capacities can generate—may restrict the range further, provided it leaves the admissible subset of $\mathcal{R}$ nonempty whenever $\mathcal{R}$ is; it is a primitive of the technology, not an object of the agent’s optimization.

Definition 6.5 (Truth-blind formulation rule). After successful production of $\mathcal{P}'_i$, let $r_i^{\mathcal{P}'_i}$ denote the deliberative record recoded to that partition. At this point the agent can compute from $r_i^{\mathcal{P}'_i}$ the same repair-consistent active set that the analyst writes as $\hat{M}_i^R(\mathcal{P}'_i, \bar{r}_i)$. A formulation rule is a possibly randomized kernel

$$\psi_i(\cdot \mid \mathcal{P}'_i, r_i^{\mathcal{P}'_i}) \in \Delta(\hat{M}_i^R(\mathcal{P}'_i, \bar{r}_i)).$$

It is record-measurable and truth-blind if its conditional distribution depends only on the displayed arguments and on internal randomness independent of the objective assignment, state draws, consequence draws, and production randomness. A truth-blind inquiry-and-formulation rule is a pair (ϕ_i, ψ_i) satisfying these restrictions. Any randomized post-refinement prior-selection rule is subject to the same record-measurability and truth-blindness requirement.

The technology is the paper’s stated representational primitive. No formal model derives a semantically new concept from nothing; the contribution is to show how experience *directs* a limited, fallible generative capacity and why that capacity does not amount to a prior over future theories. The output, and even the set of possible outputs, is not a pre-insight subjective menu: the agent cannot name a candidate refinement, assign it a probability, or condition a policy on it before production succeeds ([section 3.5](#)).

6.3 Nonidentification and the price of robustness

Theorem 6.6 (Observational nonidentification). *Fix common initial epistemic states, common record-measurable behavioral and tie-breaking rules, common truth-blind inquiry-and-formulation rules, and a finite horizon of H periods. Let $\tau \neq \tau'$ be objective assignments whose joint consequence laws agree at every state–profile pair reachable through H under either world from those initial conditions:*

$$\tau(s)(a) = \tau'(s)(a) \in \Delta(X) \quad \text{at every such } (s, a).$$

The two worlds admit a coupling under which, through H , their state draws, full consequence profiles, analyst and deliberative records, partition paths, beliefs, conjectures, standing policies, and production-and-formulation outputs coincide pathwise. If the equality holds at every pair reachable at any finite horizon, the coupling extends to the entire process.

Consequently, no truth-blind inquiry-and-formulation rule can produce an objectively correct formulation with probability one in both worlds. More generally, the same conclusion holds for any success criterion whose sets of realizable outputs are disjoint across the two worlds.

Proof. Use common state draws, common independent random seeds for production, formulation, bridge-prior selection, and tie-breaking, and an identity coupling of the joint consequence draws at every reachable pair. Proceed by induction over periods. Identical histories imply identical current partitions and standing policies, hence the same action profile at the common state. The pair is reachable, so the two objective assignments give it the same joint consequence distribution and the coupled consequence profiles coincide. Both analyst and deliberative records therefore remain identical. Record-measurability and truth-blindness then give identical belief updates, conjectures, policies, refinements, and formulations, closing the induction through H ; the all-horizons claim follows by countability.

Thus every truth-blind rule has the same output distribution in the two worlds. Because $\tau \neq \tau'$, their sets of objectively correct formulations are disjoint, so that common distribution cannot assign probability one to both sets. The same argument applies to any other disjoint pair of success sets. $\square$

Proposition 6.7 (Robust sufficiency). *A partition sufficient for every assignment in a candidate set $\mathcal{T}$ must refine $\bigvee_{\tau' \in \mathcal{T}} \mathcal{P}^{\tau'}$, where $\bigvee$ denotes the coarsest partition that refines every candidate causal-equivalence partition. A representation guaranteed to accommodate all observationally equivalent worlds is accordingly finer than any single world requires; a technology that economizes on distinctions must accept fallibility.*

Proof. Sufficiency for τ' is refinement of $\mathcal{P}^{\tau'}$ (definition 3.2); refinement of each candidate's partition is refinement of their join. $\square$

Example 1 (continued). After discovery, the record contains two labeled observations: architecture 0 objectively preferred at s_{00} , architecture 1 at s_{11} . Of the fifteen partitions of four

states, exactly four are minimally restorative—each must separate s_{00} from s_{11} : the row partition $\{\{s_{00}, s_{01}\}, \{s_{10}, s_{11}\}\}$, the column partition $\{\{s_{00}, s_{10}\}, \{s_{01}, s_{11}\}\}$, and the two partitions isolating one labeled state. The record directs inquiry (eleven partitions are excluded) without determining it (four remain). A natural complexity ordering counts the coordinates needed to describe a refinement: rows and columns are one-coordinate distinctions, the isolating partitions require two. A factor-seeking technology thus produces the row or the column refinement—and here direction runs out. Define the *column world* $\tau'(s_{rc}) = \theta^c$. Through the discovery horizon and conditional on the diagonal-state event, it agrees with the row world on the full joint mechanism at both visited states, so the two worlds generate the discovery record and every truth-blind production-and-formulation output with the same probability by the same coupling argument, conditional on the diagonal-state event. No truth-blind rule can formulate correctly in both, a randomized rule succeeds in the worst case with probability at most one-half, and by [proposition 6.7](#) a partition sufficient for both worlds must be the full partition into singletons: robustness costs every distinction the language can draw, which is exactly what a minimal technology economizes on. The example also locates the failure mode of each residual candidate: whichever factor refinement arrives, its unique consistent theory extrapolates the wrong prediction to the two unlabeled states in one of the two worlds.

7 The economics of inquiry

Whether to prosecute a question must itself be a decision, and it must be priced at exactly the moment the action-guiding Bayesian calculation is disabled: at dissonance the acting belief is undefined ([definition 5.4](#)), and the candidate answers are not represented objects ([definition 6.4](#)). This section builds the valuation from what the agent actually has—its deliberative record—and prices the rest through the success weight. The resulting statistic is a genuine estimate, not a guarantee: the agent guesses at the worth of a distinction it cannot yet name, and the model keeps that epistemic honesty rather than smuggling in the analyst’s restorative correspondence.

Assumption 7.1 (Costly, elective, dissonance-triggered production). *The technology ϕ_i operates only when agent i experiences dissonance and elects inquiry, at cost $\gamma_i > 0$ in payoff units. Let k_i count agent i ’s consecutive failed elections, reset to zero on any success. Agent i holds a delabeled achievability weight $\lambda_i^{(k)} \in (0, 1]$, nonincreasing in k with $\lambda_i^{(k)} \rightarrow 0$: a subjective probability that an election produces a refinement delivering the estimated value below, defined on that event alone and not on the identities of possible outputs. On failure or abstention the standing policy is carried forward. Awareness is cumulative: acquired distinctions are never lost, though theories formulated on them may later be rejected.*

The weight is awareness of unawareness in delabeled form—the agent can believe that inquiry may yield a usable distinction, and grow less hopeful as attempts fail, without ever describing a candidate distinction—a construct with precedent in beliefs about unpredictable

novelty (Schipper, 2025; Karni and Vierø, 2017). Its decay is the answerability meta-prior doing formal work: repeated failure is evidence that the question may be unanswerable (the record irreparable, the repertoire inadequate), and a rational inquirer eventually abandons it. That the weight prices *achievability*—not mere success—absorbs the gap between the agent’s estimate and what an arriving refinement can actually deliver, a gap that is zero under the pointwise-transparency condition below and may be positive otherwise.

Definition 7.2 (Deliberative groupings and coherent classes). A deliberative grouping G of the record r_i is a partition of its tokens. For a group $g \in G$ and observed profile a , let

$$n_g(a) = |\{t \in g : a(t) = a\}|$$

and, when $n_g(a) > 0$, let $\hat{p}_g(\cdot \mid a)$ be the empirical distribution of the consequences $x_i(t)$ among those tokens. Define

$$\Theta_i(g) = \{\theta \in \Theta : \text{TV}(\hat{p}_g(\cdot \mid a), \text{marg}_i \theta(a)) \leq \eta + w(n_g(a)) \text{ for every } a \text{ with } n_g(a) \geq \bar{n}\}.$$

The group g is coherent if $\Theta_i(g) \neq \emptyset$, and the grouping G is coherent if every one of its groups is coherent. Write $\mathcal{G}_i(r_i)$ for the finite family of coherent groupings of r_i .

Coherence, $\Theta_i(g)$, and $\mathcal{G}_i(r_i)$ depend only on the deliberative record—on observed actions and own consequences—together with the known mechanism vocabulary. They never use state labels or candidate refinements. The family $\mathcal{G}_i(r_i)$ may be empty under stochastic sampling: full support guarantees positive likelihood, not finite-sample passage of the consistency test. An empty family means only that the retained record supplies no coherent action-contingent decomposition the agent can price; it does not tell the agent whether an analyst-level restorative refinement exists. A coherent grouping is the agent’s hypothesis about which remembered particulars share a mechanism. It is not a partition of S , and the agent cannot verify before insight that any refinement of its awareness would realize it.

Definition 7.3 (Deliberative leverage). Given a coherent grouping G , let $g(t)$ be the group of token t , let $C_i(t)$ denote token t ’s cell under the current partition, and set $\sigma_i(t) := \sigma_i(C_i(t))$. The per-token robust gain of proposing action a_i for a group is

$$\Delta_i^G(t, a_i) = \min_{a_{-i} \in A_{-i}} \min_{\theta \in \Theta_i(g(t))} \left(\mathbb{E}_{x \sim \theta(a_i, a_{-i})} [\pi_i(x_i)] - \mathbb{E}_{x \sim \theta(\sigma_i(t), a_{-i})} [\pi_i(x_i)] \right),$$

the worst case over rival responses and over the mechanisms the group’s record leaves open. For $G \in \mathcal{G}_i(r_i)$, define the current-cell/group intersection

$$r_i(C, g) = \{t \in r_i : C_i(t) = C, g(t) = g\}, \quad C \in \mathcal{P}_i, \quad g \in G,$$

where empty intersections contribute zero. Define the robust fine-contingent value

$$L_i^{\text{fine}}(G; r_i) = \sum_{C \in \mathcal{P}_i} \sum_{g \in G} \max_{a_i \in A_i} \sum_{t \in r_i(C, g)} \Delta_i^G(t, a_i),$$

which lets action vary by each nonempty current-cell/group intersection, and define the current-language benchmark

$$L_i^{\text{cur}}(G; r_i) = \sum_{C \in \mathcal{P}_i} \max_{a_i \in A_i} \sum_{g \in G} \sum_{t \in r_i(C, g)} \Delta_i^G(t, a_i),$$

which requires one action for every token in a current awareness cell, although the coherent grouping is used to evaluate that action. The grouping may span more than one current awareness cell, since any future awareness partition refines rather than erases current awareness. The gross representational increment under grouping G is

$$J_i(G; r_i) = L_i^{\text{fine}}(G; r_i) - L_i^{\text{cur}}(G; r_i).$$

The deliberative leverage is

$$\widehat{B}_i(r_i) = \begin{cases} \frac{T}{|r_i|} \max(\{0\} \cup \{J_i(G; r_i) : G \in \mathcal{G}_i(r_i)\}), & \widehat{M}_i(\mathcal{P}_i, r_i) = \emptyset, \\ 0, & \widehat{M}_i(\mathcal{P}_i, r_i) \neq \emptyset. \end{cases} \quad (7)$$

The premise is that the empirical token distribution recurs, so the horizon- T estimate is T times the average per-token gross representational increment. The outside option 0 represents declining to attribute actionable value to any available grouping; it also makes leverage zero when $\mathcal{G}_i(r_i) = \emptyset$. The current-language term is a counterfactual valuation benchmark, not a new dissonant policy-switching rule: at dissonance the standing policy continues to be carried forward under [assumption 3.4](#) unless inquiry succeeds. Thus $\widehat{B}_i$ prices the incremental gross action value of making a coherent grouping actionable beyond what one action per current awareness cell can already obtain, not a complete net expected-payoff value after switching costs.

Proposition 7.4 (Participation). $\widehat{B}_i(r_i)$ is well defined, nonnegative, bounded by $T(\max \pi_i - \min \pi_i)$, and computable from the deliberative record, the standing policy, and the agent-known model primitives and test calibration. At a dissonant record, electing inquiry has represented value $\lambda_i^{(k_i)} \widehat{B}_i(r_i) - \gamma_i$, so agent i elects if and only if

$$\lambda_i^{(k_i)} \widehat{B}_i(r_i) \geq \gamma_i.$$

Moreover, at a dissonant record, $\widehat{B}_i(r_i) > 0$ if and only if some coherent grouping has positive representational increment:

$$J_i(G; r_i) > 0 \quad \text{for some } G \in \mathcal{G}_i(r_i).$$

Thus a question is priced positively only when some coherent reading of the record supports action

variation within at least one current awareness cell that robustly outperforms the best action kept uniform on that cell under the same coherent reading.

Proof. The record is finite, so it has finitely many groupings. For every coherent group, $\Theta_i(g)$ is a nonempty finite set, and all action and consequence spaces are finite. Hence every max-min expression and every $L_i^{\text{fine}}(G; r_i)$, $L_i^{\text{cur}}(G; r_i)$, and $J_i(G; r_i)$ is well defined. The set whose maximum appears in [equation \(7\)](#) always contains 0, including when $\mathcal{G}_i(r_i) = \emptyset$, proving existence and nonnegativity. For fixed C and G , put $f_{C,g}(a_i) = \sum_{t \in r_i(C,g)} \Delta_i^G(t, a_i)$. Then

$$\sum_{g \in G} \max_{a_i} f_{C,g}(a_i) \geq \max_{a_i} \sum_{g \in G} f_{C,g}(a_i),$$

because the left side permits a separate maximizing action for every intersection while the right side restricts them to a common action. Summing across current cells gives $L_i^{\text{fine}}(G; r_i) \geq L_i^{\text{cur}}(G; r_i)$ and hence $J_i(G; r_i) \geq 0$. Writing $\Delta\pi_i = \max \pi_i - \min \pi_i$, each $\Delta_i^G(t, a_i) \leq \Delta\pi_i$, so $L_i^{\text{fine}}(G; r_i) \leq |r_i| \Delta\pi_i$. For each current cell C , selecting its standing action $a_i = \sigma_i(C)$ gives zero gain for every token in that cell, so $L_i^{\text{cur}}(G; r_i) \geq 0$. Therefore $0 \leq J_i(G; r_i) \leq |r_i| \Delta\pi_i$, which gives $\widehat{B}_i(r_i) \leq T \Delta\pi_i$. Computability follows by finite enumeration using only the deliberative record, the standing policy, and the agent-known model primitives and test calibration. The election rule is the achievability-weighted represented gain net of the inquiry cost. Finally, because every $J_i(G; r_i)$ is nonnegative and the outside option is zero, the maximum in [equation \(7\)](#) is strictly positive exactly when some coherent grouping has strictly positive representational increment. $\square$

The participation rule is a coherent non-Bayesian criterion operating precisely where the acting belief is undefined, and it respects every discipline of [section 3.5](#): $\widehat{B}_i$ quantifies over the agent’s own remembered tokens and the groupings it can articulate—“situations like the ones I lost will recur, and a distinction that sorted them would let me act on them”—never over named future representations. For each coherent reading, the current-language benchmark first values the best action already expressible uniformly on each current awareness cell; gross representational leverage is only the additional robust gain from conditioning action on the finer grouping. The estimate can be wrong, and honestly so: the agent does not know whether a distinction realizing its best grouping exists, which is why $\lambda_i^{(k)}$ prices achievability. Two comparative properties organize the applications. *Action separation, not information*: the value of a question tracks whether a finer distinction could improve action beyond the best current-cell-uniform response, not how much residual uncertainty remains; an agent for whom no coherent grouping yields positive representational increment declines to inquire. *Rational cessation*: an agent for whose record no coherent grouping yields positive representational increment has $\widehat{B}_i = 0$, so sustained success can extinguish inquiry even though awareness stays incomplete; and, by the decay of $\lambda_i^{(k)}$, so can sustained failure—a true, valuable, repairable question can be rationally abandoned.

Remark 7.5 (Enlightened inertia). *The model deliberately treats acquiring a distinction and acting on it as sequential decisions. Pre-insight the agent uses $\hat{B}_i$ as a gross estimate of representational leverage, while the known switching hurdle κ_i is applied only after a concrete refinement, belief, conjecture, and policy make the realized horizon gain well defined (assumption 3.4 and proposition 7.4). This timing is a bounded-rationality modeling convention, not a logical implication of unawareness. Once inquiry delivers a concrete refinement, the realized gain may fail to clear κ_i , so the agent carries its old policy forward despite knowing that its earlier causal account was incomplete. Away from pointwise transparency, that enlightened inertia can be unanticipated because pre-insight leverage is only an estimate and the realized refinement may deliver a different gain. Under pointwise transparency, by contrast, exact and repair-invariant leverage can make later enlightened inertia foreseeable; the model nevertheless preserves the two-stage timing as a deliberate simplification. Enlightened inertia remains a second source of persistence, distinct from an unpriced question, whether or not mechanisms are deterministic, and it feeds the performance differences of section 9.*

Remark 7.6 (What determinism certifies—and what it does not). *Determinism supplies a limited but useful separability certificate. Two tokens sharing the same observed pair (C, a) but differing in consequence cannot have arisen from the same state, because a state and action profile fix the consequence. Any restorative refinement must therefore separate such a certified conflict. Determinism alone does not, however, make every deliberative grouping realizable. Tokens observed under different action profiles can be grouped coherently even when their underlying particulars cannot be separated in the proposed way. In general, $\hat{B}_i$ therefore remains a deliberative estimate even with deterministic mechanisms.*

A stronger, record-level condition restores exactness. In the deterministic zero-tolerance case with $\bar{n} = 1$ and $w \equiv 0$, call a record pointwise transparent if $\Theta_i(\{t\})$ is a singleton for every retained token t . Every token pins one mechanism, and every restorative refinement separates tokens carrying incompatible pinned mechanisms. A fine-contingent proposal chooses actions separately at each current-cell/pinned-group intersection, so it is implementable under any restorative refinement; a coherent group may span current cells without merging them because every new partition refines the current one. The current-language benchmark, which chooses one action per current cell, is already implementable. Consequently both value terms, and therefore their representational increment, are exact and repair-invariant. The running example’s differentiated discovery record is pointwise transparent, so its leverage calculation has precisely this property.

Under stochastic mechanisms, differing consequences at one (C, a) -pair may reflect mechanism noise rather than distinct states, and even a coherent grouping need not be realizable. There $\hat{B}_i$ is genuinely an estimate, and the achievability weight prices the possibility that no arriving distinction delivers it. Counterfactual transparency—a small $\Theta_i(g)$, ideally a singleton—makes the estimate sharper; opacity enlarges $\Theta_i(g)$, lowers robust leverage, and can rationally suppress inquiry.

Example 1 (continued). At the discovery record ($T = 2$, diagonal states served), each firm holds two tokens sharing the profile $(0, 1)$ but differing in payoff: firm 1 won at s_{00} and lost at s_{11} , firm 2 the reverse. Because mechanisms are deterministic, the conflicting payoffs *certify* the two tokens

as distinct states (remark 7.6), and the agent’s best coherent grouping splits them: on the winning token the fine optimal gain is zero, while on the losing token switching to the pinned architecture gains $\frac{1}{2}$ against every rival action—a differentiated loss becomes a pooled half, or a pooled half a differentiated win. Thus $L_i^{\text{fine}} = \frac{1}{2}$. Under the coarse one-cell current language, keeping the standing action gives aggregate gain zero, while switching uniformly gains $\frac{1}{2}$ on the losing token and loses $\frac{1}{2}$ on the winning token, so $L_i^{\text{cur}} = 0$ and $J_i = \frac{1}{2}$. Consequently $\hat{B}_i = (2/2)(\frac{1}{2}) = \frac{1}{2}$ for both firms—identical gross representational leverage from opposite experiences—so each elects iff $\lambda_i^{(0)} \hat{B}_i \geq \gamma_i$, i.e. $\lambda_i^{(0)}/2 \geq \gamma_i$. The positive value is genuinely representational because no uniform current-language action improves on the standing policy. Acting on the row refinement, once acquired, means re-differentiating by row at the switching hurdle κ_i ; the post-refinement horizon gain remains exactly $\frac{1}{2}$, so the firm re-differentiates when $\kappa_i < \frac{1}{2}$ and—knowing the row law yet finding the change not worth its cost—holds its old policy when $\kappa_i \geq \frac{1}{2}$, the deterministic face of enlightened inertia (remark 7.5). Under the model’s two-stage timing it may still elect inquiry when the participation condition holds; in this transparent case, a hurdle at or above the exact gain makes later enlightened inertia foreseeable, a consequence of the deliberate timing convention rather than of unawareness. The valuation used no conjecture about the rival and no reference to which refinement might arrive; transparency (winner-take-all) makes each $\Theta_i(g)$ a singleton, so the deliberative estimate is exact. For rational cessation and its converse, consider a later stretch that serves the two off-diagonal states, by which point firm 1 has acquired the row refinement (certain of the true assignment) while firm 2 has acquired the column refinement (certain of the false one): their policies differ at those states, firm 1 wins both, and the labels revealed there falsify firm 2’s column theory. For firm 1, both the fine and current-language values are zero, so $L_1^{\text{fine}} = L_1^{\text{cur}} = J_1 = \hat{B}_1 = 0$: the firm holding the truth rationally never inquires again, and its awareness stays permanently incomplete. For firm 2, two tokens disagree with its column policy and contribute $\frac{1}{2}$ each under fine contingency, while two agree and contribute zero, so $L_2^{\text{fine}} = 1$. Within either current column, reversing the action gains $\frac{1}{2}$ on one token and loses $\frac{1}{2}$ on the other, so $L_2^{\text{cur}} = 0$, $J_2 = 1$, and $\hat{B}_2 = (2/4)1 = \frac{1}{2}$. Defeat thus concentrates the evidence, dissonance, and incentive in the losing firm in this example, which elects whenever $\lambda_2^{(k_2)} \hat{B}_2 \geq \gamma_2$ and, upon success, reaches complete awareness in one step (both column cells must split). The winner/loser asymmetry— $\hat{B}_1 = 0 < \hat{B}_2$ —is the durable point; the appendix records the exact value of $\hat{B}_2$.

8 The pivot: formulation, warrant, and the return to Bayes

Successful production yields a refinement $\mathcal{P}'_i$; the truth-blind formulation rule selects $m' \in \hat{M}_i^R(\mathcal{P}'_i, \bar{r}_i)$ from the repair-consistent theories on the recoded record; and the agent forms a full-support *bridge prior* $\mu_i^+ \in \Delta(M(\mathcal{P}'_i))$, from which ordinary updating resumes. Nothing in Bayesian coherence determines μ_i^+ from the pre-dissonance belief—the domains differ—and this

nonuniqueness is not a defect to be repaired but the boundary between insight and judgment. Restricting and renormalizing μ_i^+ on $\hat{M}_i^R(\mathcal{P}'_i, \bar{r}_i)$ yields the initial acting belief on the refined language, and the resulting probability assigned to the formulated theory is what we call *representation-relative coherence*. The question of this section is how much of that confidence the evidence actually licenses.

The conjecture on the refined partition is initialized from the recoded empirical frequencies of observed rival actions wherever the record contains observations, with a fixed default on previously unvisited cells, and is updated empirically thereafter. Thus both inputs needed for the next standing policy—the bridge belief and the strategic conjecture—are defined on the expanded language only after the refinement has arrived.

Definition 8.1 (Reflective assessment, warrant, and the gap). *A reflective assessment problem is a pair $(\mathcal{T}, \nu)$: a finite set of candidate objective assignments and a full-support reflective prior. It is an analyst-level construct: conditioning on the events below uses the analyst record and the inquiry history. Let h_i denote the pre-production inquiry history, including earlier elections and failures. Given $(\bar{r}_i, h_i)$ together with a realized successful production-and-formulation output $(\mathcal{P}'_i, m')$, the selection-adjusted warrant W of $(\mathcal{P}'_i, m')$ is the ν -posterior probability that m' is objectively correct—that $\tau'(s) = m'(C_{\mathcal{P}'_i}(s))$ for all s —conditional on that record and history and on the realized output. The warrant gap is the representation-relative coherence probability of m' minus its warrant W .*

Theorem 8.2 (The insight is not evidence). *Condition on the analyst record $\bar{r}_i$ and the pre-production inquiry history h_i . Under a record-measurable, truth-blind inquiry-and-formulation rule, conditioning additionally on a particular successful output $(\mathcal{P}'_i, m')$ does not change reflective odds:*

$$W = \nu(m' \text{ is objectively correct} \mid \bar{r}_i, h_i) .$$

Post-restoration fit is guaranteed by the restoration property, so passing the repair test carries no information beyond the record either. More generally, among candidates in $\mathcal{T}$ that induce the same likelihood for $(\bar{r}_i, h_i)$, reflective prior odds are preserved exactly.

Proof. For $\tau' \in \mathcal{T}$, the posterior conditional on the record, history, and realized output is proportional to

$$\nu(\tau') \Pr(\bar{r}_i, h_i) K_i(\mathcal{P}'_i, m' \mid \mathcal{P}_i, \bar{r}_i, h_i) ,$$

where K_i is the conditional kernel generated by election, production, and formulation. Election is determined by the deliberative record and failure counter, while production and formulation are truth-blind. Consequently the last factor depends on the displayed record and history but not on τ' , and cancels from posterior odds. Passing the repair test is a deterministic implication of the record and realized output and therefore contributes no additional likelihood factor. If the record-and-history likelihoods are also equal across two candidates, their posterior odds equal their prior odds. $\square$

The theorem locates the evidential boundary: everything evidential is in the record; nothing evidential is in the arrival of the insight or in its retrospective success at organizing the experience that produced it. The conditioning must be at the analyst level, because the election and the success set are functions of the state-labeled record; were one to condition only on the deliberative record, the distribution of the hidden states would itself be τ' -dependent and the cancellation would fail—the insight would spuriously appear to be evidence. The practical force is a prohibition on double counting: the same experience that directed production cannot also confirm the resulting theory, because any restorative output fits it by construction. Truth blindness is the polar case; were production correlated with the objective assignment—social transmission from an informed source—the arrival would itself be evidence, and the adjustment could move confidence in either direction.

Corollary 8.3 (Self-audit requires sufficiency for the candidates). *The assessment $(\mathcal{T}, \nu)$ is expressible by agent i only if $\mathcal{P}_i$ is sufficient for every $\tau' \in \mathcal{T}$, hence only if $\mathcal{P}_i$ refines the join of [proposition 6.7](#). An agent holding a minimal restoration therefore generally cannot formulate the comparison that measures its own warrant gap: the overconfidence is invisible from within the representation that produces it. Short of that refinement, warrant must be supplied externally—by fresh evidence whose likelihoods differ across candidates, by an assessor with broader awareness, or by a future self.*

Proof. Entertaining τ' as a hypothesis requires expressing it, which is sufficiency ([definition 3.2](#)); sufficiency for every candidate is refinement of the join. $\square$

The pivot is now complete, and it is the paper's fulcrum. Dissonance broke the agent's Bayesian competence; inquiry produced a distinction; formulation and the bridge prior rebuilt a probability space; and judgment—fresh evidence with differential likelihoods, honestly accounted—resumes as ordinary updating on the expanded language. The agent has returned to the Bayesian world carrying a warrant gap of known location and, by [corollary 8.3](#), generally unknowable (to it) size.

Example 1 (continued). Suppose firm i 's technology produces the column refinement. The recoded record pins the unique consistent theory: architecture 0 on the first column, 1 on the second. Representation-relative coherence assigns it probability one. Now conduct the reflective assessment with $\mathcal{T} = \{\text{row world, column world}\}$ and a symmetric prior. The two worlds assign the realized diagonal-only record equal probability by the same coupling argument conditional on the diagonal-state event, and the technology is truth-blind, so by [theorem 8.2](#) the reflective posterior equals the prior: $W = \frac{1}{2}$, and the warrant gap is exactly one-half. The same gap afflicts the firm that acquires the *row* refinement: its coherence in the true assignment is also probability one, and its warrant is also one-half. Unwarranted certainty is not a symptom of error—early confidence outruns the evidence even when it happens to be right, and what separates the two firms is the objective law, which neither observes. By [corollary 8.3](#), neither firm can conduct this assessment itself: expressing both candidate worlds requires the full partition

into singletons. One fresh differentiated observation at an off-diagonal state closes the gap completely—the two worlds predict opposite labels there—while pooled observations, in any number, close nothing. Warrant, like leverage, is purchased with differentiated action.

9 Strategic dynamics: when everyone lives in a Bayesian world

The chain now runs inside a population. Theories determine standing policies; policies determine which conditionals are visited; visited conditionals determine dissonance and leverage; and refinements feed back into theories. This section characterizes where the process comes to rest. The object of interest is not a strategic equilibrium but an epistemic condition: a configuration of the population in which *every agent permanently inhabits a Bayesian world*—models fit, questions are dead, and no represented calculation prices further inquiry above its cost.

Definition 9.1 (Self-confirming unawareness equilibrium). *A profile of epistemic states $\xi_i = (\mathcal{P}_i, r_i, \bar{\mu}_i, q_i, \sigma_i, k_i)$, $i \in N$ —awareness, retained record, reference posterior, conjecture, standing policy, and failure counter, with acting beliefs μ_i derived by judged updating—is a self-confirming unawareness equilibrium (SCUE) if, under the play it induces,*

- (i) *each standing policy is stationary under the switching rule of [assumption 3.4](#): no switch clears the hurdle, $T[U_i(\sigma_i^* \mid \mu_i, q_i) - U_i(\sigma_i \mid \mu_i, q_i)] \leq \kappa_i$, which is subjective optimality itself when $\kappa_i = 0$;*
- (ii) *each conjecture q_i agrees, in the limit of its empirical frequencies, with the play induced by σ_{-i} at every on-path cell;*
- (iii) *no dissonance survives: each agent's active set, computed from its record as play accumulates, is almost surely eventually nonempty, and the acting-belief mass on any η -inadequate theory vanishes almost surely; and*
- (iv) *no agent elects inquiry: $\lambda_i^{(k_i)} \widehat{B}_i(r_i) < \gamma_i$ along the induced records, which holds in particular whenever $\widehat{B}_i = 0$.*

The record and failure counter belong in the state: dissonance status, leverage, and the achievability weight are functions of retained evidence and inquiry history, so two histories with identical partitions, beliefs, and policies but different records can differ in equilibrium status. Conditions (ii)–(iii) are stated asymptotically because empirical conjectures and stochastic tests fluctuate at finite samples. Deterministic consequence mechanisms make a contradictory test observation irreversible, but they do not eliminate sampling variation in states or in rivals' actions within an agent's awareness cell; empirical conjectures therefore generally still converge only asymptotically.

Conditions (i)–(ii) are subjective rationality with confirmed conjectures: the self-confirming conditions of the fixed-language tradition (Fudenberg and Levine, 1993; Kalai and Lehrer, 1993b).

Conditions (iii)–(iv) are the new, epistemic ones: the language survives its own evidence, and the question has no priced value. A SCUE in which every partition is sufficient for τ and beliefs concentrate near the truth is simply a self-confirming equilibrium; the concept’s interest lies entirely in the other cases.

Definition 9.2 (Dissonant stasis). *Agent i is in dissonant stasis from some period on if dissonance recurs while it never both elects and succeeds—because $\lambda_i^{(k_i)} \widehat{B}_i(r_i) < \gamma_i$ (no priced repair: zero leverage, or an abandoned question after the achievability weight has decayed, possibly at an irreparable record), or because elected production fails at every subsequent attempt. Its standing policy is carried forward with a standing anomaly. A population profile is absorbing if every agent is, from some period on, either in a SCUE role (never dissonant) or in dissonant stasis.*

Remark 9.3 (The grain of truth, restored endogenously). *Rational-learning convergence requires a grain of truth: the environment absolutely continuous with respect to beliefs (Kalai and Lehrer, 1993a). Under unawareness the condition can fail at the level of language—the truth may be inexpressible—and statistical dissonance is precisely the detector of that failure. Each successful refinement restores the grain on the path of play: post-restoration, some expressible theory matches the visited conditionals within tolerance. A SCUE is a configuration in which the grain of truth holds on-path for every agent and no one has a priced reason to look for the off-path remainder.*

Assumption 9.4 (Regularity). (i) *Strict adequacy. At each visited pair every theory is either η -adequate or strictly inadequate—none sits exactly at the adequacy boundary—so lemma 5.2 applies for every agent, theory, partition profile, and policy profile. In the deterministic case $\eta = 0$ this holds automatically and the sequential detector recovers the exact test.* (ii) *Production randomness. The randomization in each ϕ_i is independent across elections and of the state and consequence draws, and, conditional on an election at a repairable record ($\mathcal{R}(\mathcal{P}_i, \bar{r}_i) \neq \emptyset$), production succeeds with probability at least $\underline{\pi} > 0$. At an irreparable record production fails surely (definition 6.4).*

Part (i) is the genericity that makes each theory’s evidence process settle cleanly relative to the capacity threshold (lemma 5.2); it rules out only a boundary knife-edge and holds automatically in the deterministic case. Part (ii) gives every elected repairable question a real chance of resolution; finiteness of elections comes instead from the decay of the achievability weight, so no separate never-produce exclusion is needed. The agent, unable to tell a repairable record from an irreparable one, may elect either; the two are distinguished only in their success probabilities, not in the agent’s decision.

Theorem 9.5 (Epistemic absorption). *Under assumption 9.4, almost surely:*

- (a) *at most $n(|S| - 1)$ refinements occur, and every agent’s awareness partition is eventually constant;*
- (b) *only finitely many inquiry elections occur—each agent stops electing once its achievability weight has decayed enough that $\lambda_i^{(k)} T(\max \pi_i - \min \pi_i) < \gamma_i$, and this happens after finitely many consecutive failures because $\widehat{B}_i$ is bounded and $\gamma_i > 0$; and*

- (c) *there is an almost surely finite period after which no agent's language changes and no agent pays an inquiry cost: every agent's epistemic life thereafter is judged-Bayesian updating within a fixed language, punctuated at most by belief-driven policy switches under [assumption 3.4](#) and by dissonance events that fail the participation threshold.*

Moreover, on the event that the standing profile is eventually constant,

- (d) *every agent's active set is eventually constant; every theory retained in that eventual active set is η -adequate at every recurrent pair under terminal play, every theory η -inadequate at a recurrent pair under terminal play is eventually absent, and each agent is eventually either never dissonant—it lives in a Bayesian world—or in dissonant stasis ([definition 9.2](#)).*

The proof is in [appendix B](#).

Proposition 9.6 (Rest points and persistence).

- (i) *Suppose awareness partitions and standing policies are eventually constant almost surely, conjectures and reference posteriors converge, and each non-stasis agent's standing policy is a strict no-switch point at the limit—either its switching gain is bounded strictly below κ_i , or, when $\kappa_i = 0$, the standing action is the unique U_i -maximizer so the gain is exactly zero. Then the limiting configuration is a partial SCUE: the SCUE conditions hold for every non-stasis agent, taking the stasis agents' constant carried-forward policies as fixed environment. (Belief coordinates need not literally freeze; the strict slack transfers the no-switch condition to the limit by continuity of the finite-dimensional payoff comparison.)*
- (ii) *Conversely, consider a configuration whose epistemic states satisfy the SCUE conditions, whose play generates only records with $\widehat{B}_i = 0$, and in which each agent's standing action at each cell is strictly optimal under every conjecture and every theory in its active set.*
- (a) *With deterministic mechanisms, the configuration persists: dissonance, elections, and policy changes never occur, and the SCUE conditions continue to hold, with conjectures converging empirically to induced play.*
- (b) *With stochastic mechanisms, suppose in addition that, at the entry configuration, every agent's active set is nonempty, every theory it contains is η -adequate under the configuration's play and carries no accumulated evidence against it ($E_i^m \leq 1$, automatic when the configuration is entered on a freshly refined language), and the standing action is strictly optimal under every conjecture and every η -adequate theory. Then, conditional on the entry history, with probability at least $1 - \sum_i e^{-b_i}$, no agent is ever dissonant, no switch or election occurs, and the SCUE conditions hold in every subsequent period, with conjectures converging empirically to induced play. The residual $\sum_i e^{-b_i}$ is the sequential test's false-alarm bound: it shrinks as dissonance capacities rise, and vanishes in the deterministic case—so in stochastic environments capacity governs the robustness of a rest point.*

The proof is in [appendix B](#).

Remark 9.7 (What the theorem does and does not claim). *Theorem 9.5 is a statement about the epistemic process—the death of inquiry—not about the convergence of play. Within a Bayesian world, standing policies may in principle continue to cycle as beliefs and conjectures chase one another; that is the classical convergence problem of learning in games (Fudenberg and Levine, 1993; Kalai and Lehrer, 1993a), it is orthogonal to this paper’s contribution, and the theorem deliberately does not resolve it. Part (d) and the rest-point characterization are accordingly conditional on play stabilizing, which in every configuration featured in this paper is immediate: their beliefs make one action per cell strictly dominant. Switching costs temper the cycling: the inaction band absorbs belief fluctuations too small to be worth acting on, so only swings whose horizon-T gain exceeds κ_i move play—a stabilizing force, though not one that rules out large-amplitude cycles. The decay of $\lambda_i^{(k)}$ is what makes inquiry die out even when the question is genuine: an agent facing a true, valuable, but hard-to-repair defect abandons it once repeated failure has eroded its hope. Abandoned questions are thus a substantive absorbing outcome, not an artifact—inquiry ceases because it is answered, because it is unpriced, or because it is given up. Were $\lambda_i^{(k)}$ held fixed instead of decaying, an agent could elect a doomed inquiry forever, paying γ_i each period for a repair that never arrives; the decay rules this out and is the model’s account of when a rational inquirer stops.*

Remark 9.8 (Rational anomaly tolerance). *Zero-leverage dissonant stasis is a recognizable condition, not a pathology of the model: the agent registers that its causal account is broken—its test rejects every expressible theory—yet its deliberative valuation finds no coherent grouping with positive representational increment, so repairing the account is worth nothing to it. Standing anomalies tolerated by working scientists and managers have exactly this structure, and the model locates them: anomaly tolerance is rational precisely when the agent’s coherent groupings carry no gross representational leverage. Two sharper instances complete the taxonomy. At an irreparable record ([remark 6.3](#)) no expressible repair exists at any price—the anomaly is not merely unprofitable but impossible to fix within the current grammar—though the agent, unable to certify irreparability under noise, may keep electing while its deliberative record supports positive leverage and until the achievability weight decays. And an abandoned question is a repairable, positively-valued defect the agent stops pursuing because repeated failure has driven $\lambda_i^{(k)} \hat{B}_i$ below cost: rational surrender, not indifference.*

Proposition 9.9 (On-path adequacy, and where error lives). *In a SCUE, acting-belief mass on every η -inadequate theory vanishes almost surely. Under the strict-adequacy condition of [lemma 5.2](#), or in its exact deterministic case, every theory remaining in the eventual support of μ_i is η -adequate at every on-path pair. All falsity retaining nonvanishing acting-belief mass is confined to what the configuration does not expose: mechanism assignments at states pooled with causally different states whose mixture remains within tolerance of a single mechanism at the visited profiles, and counterfactual claims at unvisited profiles.*

Proof. The first claim is [definition 9.1\(iii\)](#). Under strict adequacy, or under the lemma’s exact

deterministic argument, lemma 5.2 makes the active set eventually constant and excludes every η -inadequate theory; judged updating restricts the acting belief to that set. The residual clause is the complement of the on-path constraints. $\square$

Proposition 9.10 (Capacity: detection speed and robustness). *Fix agent i and stochastic mechanisms satisfying assumption 5.3.*

- (i) *Detection. Fix an exposed standing profile and suppose the evidence processes are freshly initialized when that profile begins. For each theory m , let d_i^m be the supremum of the positive per-period expected log-growth rates available from the fixed betting components associated with its recurrent inadequate pairs in appendix A, and define $D_i = \min_m d_i^m > 0$. The expected number of periods until agent i 's dissonance satisfies $\mathbb{E}[\tau_{\text{dissonance}}] = O(b_i/D_i)$ and $\mathbb{E}[\tau_{\text{dissonance}}] = \Omega(b_i)$.*
- (ii) *Robustness. At a non-exposed stochastic profile entered under the hypotheses of proposition 9.6(ii-b), agent i spuriously becomes dissonant with probability at most e^{-b_i} .*

In stochastic environments, b_i creates a detection–robustness tradeoff: a larger b_i lowers false-alarm risk but requires more accumulated evidence to detect a broken language. Under deterministic $\eta = 0$ mechanisms, a recurrently inadequate theory is instead rejected at its first diagnostic impossible consequence, so detection speed is governed by the occurrence probability of diagnostic evidence rather than by b_i ; adequate theories have zero false-alarm probability, and capacity affects neither detection nor robustness in the exact deterministic model.

Proof. (i) Exposure makes $d_i^m > 0$ for every expressible theory and finiteness makes $D_i > 0$. Because the countable betting support need not attain the supremum, choose for each m a recurrent inadequate pair j_m and a positive-weight component W_m at that pair whose per-period expected log-growth is at least $d_i^m/2$, and let π_m be its within-pair mixture weight. Component domination gives $E_i^m(t) \geq v_{j_m} \pi_m W_m(t)$ for fixed positive weights $v_{j_m} \pi_m$, so the rejection time τ_m is no later than the crossing time of $\log W_m$ above $b_i - \log(v_{j_m} \pi_m)$. The stochastic construction has bounded log-increments; standard bounded-increment stopping-time and Wald reasoning therefore gives $\mathbb{E}[\tau_m] \leq 2(b_i + c_m)/d_i^m$, where c_m is independent of b_i . Since $\tau_{\text{dissonance}} = \max_m \tau_m \leq \sum_m \tau_m$ and the theory set is finite, $\mathbb{E}[\tau_{\text{dissonance}}] = O(b_i/D_i)$ with constants independent of b_i . Conversely, bounded one-period growth of each freshly initialized stochastic evidence process implies that no process can cross e^{b_i} before a number of periods proportional to b_i , yielding $\mathbb{E}[\tau_{\text{dissonance}}] = \Omega(b_i)$. Part (ii) is the per-agent bound of proposition 9.6(ii-b); neither rate argument applies to the separate exact deterministic detector. $\square$

Proposition 9.11 (Persistent performance differences). *There exist primitives and absorbing configurations in which agents hold different terminal partitions and theories, and expected per-period payoffs differ strictly and permanently, with no disadvantaged agent closing the gap—because for each the*

corrective process has died economically at one of its two stages: either its question is unpriced (dissonant stasis, $\lambda_i^{(k_i)} \widehat{B}_i(r_i) < \gamma_i$ from some period on), or, having asked and refined, the policy change its new model recommends is priced below the switching hurdle (enlightened inertia, T -horizon gain $\leq \kappa_i$, [remark 7.5](#)).

The stasis channel is established by the instance below; the enlightened-inertia channel is the deterministic possibility of [remark 7.5](#) (a firm that acquires the true distinction yet, facing κ_i above the gain, forgoes the improvement its rival captures). The mechanism deserves emphasis. Performance differences persist not because an equilibrium concept freezes them but because the process that would erase them has died economically: the disadvantaged agent's question is priced below the cost of asking—through zero leverage, an eroded achievability weight, or an irreparable record—or, having asked and learned, it finds the switch its new model recommends priced below the hurdle and holds its old policy ([remark 7.5](#)). Terminal heterogeneity in awareness is path dependence in the strong sense—agents with identical preferences and identical Bayesian competence, facing the same objective system, end at different languages and different payoffs because their histories posed different questions.

Remark 9.12 (Two sources of heterogeneity). *Terminal heterogeneity has two independent origins here. Agents identical in preferences, competence, capacity, and cost can still end at different languages and payoffs, because their histories posed different questions and their truth-blind technologies produced different repairs—path dependence that begins before probability. Agents identical in everything but their dissonance capacity b_i and switching hurdle κ_i also diverge: in stochastic environments capacity affects detection speed and the robustness of rest points ([proposition 9.10](#)), while the hurdle sets the propensity to enlightened inertia ([remark 7.5](#)); the capacity channel disappears under exact deterministic contradiction. The first is the awareness-path microfoundation; the second a parametric cognitive-style one—and neither is available in the fixed-language tradition.*

Definition 9.13 (Diagnostic environments). *An expressible theory is payoff-relevantly false for agent i under a profile if the truth would render i 's standing action suboptimal at some visited state—not merely shrink the set of optimal actions. The primitives $(\Theta, \tau, \rho, \pi)$ are diagnostic if, under every profile sustainable in a SCUE, every payoff-relevantly false theory is η -inadequate.*

Proposition 9.14 (No payoff-relevant error in diagnostic environments). *In a diagnostic environment, every SCUE is free of payoff-relevant error. Payoff-relevant falsity survives only in environments that admit concealing profiles, and it requires them to be sustained: collective error needs the absence of both corrective forces, strategic variety inside agents' categories and environmental punishment of uniform misservice.*

Proof. By [definition 9.1\(iii\)](#), theories surviving in a SCUE are not η -inadequate; in a diagnostic environment every payoff-relevantly false theory is. The second claim restates [definition 5.5](#): a false theory survives only under concealing profiles, and, under its no-legacy proviso, [proposition 5.6](#) makes concealment necessary for survival and sufficient for avoiding eventual dissonance. $\square$

Proposition 9.15 (Diagnosticity is robust). *Because an agent's consequence x_i carries its realized payoff, an objectively suboptimal standing action is exposed the moment it is played—unless its on-path consequence happens to match what the agent's theories predict. Call the primitives strictly punishing if, under every SCUE-sustainable profile, whenever an agent's standing action is objectively suboptimal at a visited state, the on-path conditional at that visited pair lies more than η in total variation from every active theory's prediction. Strictly punishing primitives are diagnostic, so by [proposition 9.14](#) their every SCUE is free of payoff-relevant error. The condition is a finite conjunction of strict inequalities, hence open in (Θ, τ, π) : diagnosticity is robust, surviving small perturbations of the primitives. Concealment is open too when $\eta > 0$ —a theory predicting within η of the on-path conditional tolerates perturbation—and collapses to the exact-match knife-edge only as $\eta \rightarrow 0$, where the running example lives ($\beta = 1$).*

Proof. If an agent's standing action is objectively suboptimal at a visited state, strict punishment places the on-path conditional there more than η from every active theory's prediction, so every active theory is η -inadequate at that pair and the active set empties—dissonance, not a SCUE. Hence in a SCUE no standing action is payoff-relevantly false, which is the diagnostic property. Openness is the finite conjunction of strict inequalities; for $\eta > 0$ the concealing region (predictions within η of the on-path conditional) is open as well, so only at $\eta = 0$ does concealment reduce to the exact-match knife-edge ($\beta = 1$ in the running example). $\square$

Example 1 (continued). *The trap.* Set $\beta = 1$ and suppose both firms acquired the column refinement after discovery and became certain of the false column assignment. Each plays the column policy; the firms pool at every state; every payoff is one-half; and one-half is exactly what the column theory predicts under pooled play. The configuration is a SCUE with payoff-relevant error: policies are subjectively optimal, conjectures confirmed, and no dissonance ever occurs because the profile is concealing—so the technology never triggers, whatever its cost or reliability. The warrant gap of [section 8](#) is permanent: unbounded experience accumulates while each firm remains exactly half-warranted, the missing half being the reflective weight on the world their shared language cannot express.

The knife edge. Now let $\beta < 1$. In the first period serving an off-diagonal state, the pooled firms receive $\beta/2$ where their theory predicts one-half: the demand side reveals the label, dissonance returns, $\hat{B}$ turns positive, and the reflective posterior becomes degenerate. Payoff-relevant self-confirmation is a knife-edge property of perfectly forgiving demand, and the destroyed value $1 - \beta$ at each missed period is itself the corrective signal: what demand withholds from the firms, their epistemics receives as evidence.

Persistence. Run the asymmetric configuration at $\beta < 1$: firm 1 holds the row refinement and the true assignment; firm 2 remains coarse and, by $\gamma_2 > \lambda_2^{(k_2)} \hat{B}_2$, sits in dissonant stasis. Here firm 2's coarse record genuinely rejects all three constant theories—its differentiated losses defeat the constant theory matching its own action, its pooled payoff of one-half defeats the opposite theory's $\beta/2$ prediction, and neutrality fails at the differentiated losses—so it is dissonant rather than merely mistaken, and it stays put because its question is priced below cost. With constant

action c , the two correctly served tokens contribute zero and the two missed tokens contribute $\frac{1}{2}$ each under fine contingency, so $L_2^{\text{fine}} = 1$. A uniform current-language reversal has aggregate robust gain $\beta - 1 < 0$, while retaining the standing action gives zero, so $L_2^{\text{cur}} = 0$, $J_2 = 1$, and, for this balanced four-token record with $T = 2$, $\hat{B}_2 = \frac{1}{2}$. For other realized record compositions the gross representational leverage remains positive when some coherent grouping supplies a positive increment and is bounded as in [proposition 7.4](#). Firm 1 offers the preferred architecture in the row its rival misses, winning both those states and splitting the rest, so under uniform ρ the firms earn three-quarters and one-quarter, permanently. (At $\beta = 1$ this instance instead sends firm 2 oscillating: whichever constant architecture it plays, the surviving constant theory prescribes the other, so its policy never settles—an instructive contrast, but not a persistent *stable* difference, which is why the diagnostic environment $\beta < 1$ is the right home for the result.) Every route out—strategic variety, demand harshness, a cheaper or more hopeful inquiry—is exactly one of the model’s named forces.

10 Discussion

10.1 Implications for the theory-based view

The model supplies the theory-based view with the object its formal treatments presuppose and, in doing so, changes what several of the view’s claims mean.

Strategic variety has a pre-probabilistic source. In the model, agents with identical preferences, identical Bayesian competence, and identical objective opportunities diverge because their experiences posed different questions, their technologies produced different repairs, and their positions extinguished inquiry at different points. Heterogeneous prior understanding generates heterogeneous questions; questions generate heterogeneous languages; languages generate heterogeneous theories, plans, and payoffs. This sequence begins before probability does, and it is path-dependent in the strong sense: the order of experiences shapes the terminal language, and agents end at different languages and different payoffs ([proposition 9.11](#)).

The value of experimentation is not information. An agent prices a question by whether some coherent reading of its own record supports a robust representational increment beyond the best action already uniform on each current awareness cell ([proposition 7.4](#)), not by how much its resolution would reduce uncertainty. This reframes theory testing: high-entropy uncertainty with no gross representational leverage is rationally ignored, and narrow contrasts that straddle an action threshold are rationally pursued. It also explains rational cessation—why successful firms stop asking—the loser’s greater incentive to inquire, since competitive defeat concentrates the evidence and the dissonance in the defeated firm, and the abandonment of genuine but stubborn questions, as the achievability weight decays under repeated failure.

Early confidence outruns evidence even when correct. The warrant results discipline a claim at the center of the empirical program on scientific entrepreneurship ([Camuffo et al., 2020](#);

Zellweger and Zenger, 2023). A founder's theory that beautifully organizes the experiences that generated it has, by [theorem 8.2](#), earned nothing by that organization: the fit was guaranteed by construction. Validation requires evidence whose likelihood differs across the candidate worlds—and, by [corollary 8.3](#), the founder typically cannot even express the comparison, which is a formal argument for outside challenge (boards, investors, advisors with different awareness) as a warrant technology rather than a monitoring technology.

Markets discipline theories through two channels, and their absence is predictable. Payoff-relevant collective error requires both concealment by play and forgiveness by demand ([proposition 9.14](#)). Industries where customers absorb misservice—captive buyers, no substitutes, third parties bearing the cost—can sustain shared false theories indefinitely; industries with strong outside options discipline theories even under consensus. Where demand is forgiving, the only corrective is strategic variety in beliefs, which gives belief heterogeneity a social value distinct from its private value: the entrant with a deviant theory is the industry's epistemic infrastructure.

Anomaly tolerance is an absorbing-state phenomenon. Firms that register their causal account is broken and rationally decline to fix it ([remark 9.8](#)) are not failing at learning; their standing anomalies carry no gross representational leverage, or have grown too costly to keep pursuing. Such firms sit in dissonant stasis, outside any Bayesian world, which is why the condition is an absorbing state rather than an equilibrium. The model locates where tolerated anomalies accumulate: where consequences are opaque, so the agent cannot price the forgone value of a repair, and—even under transparency—near decisions whose robust optimum is insensitive to the unresolved structure.

10.2 The cognitive interpretation

The model's architecture tracks a classical account of inquiry ([Loneragan, 1992](#); [Ryall and Wilkins, 2018](#)), and the correspondence is disciplining rather than decorative: it dictates which formal objects must not be conflated. Experience is the flow of periods; the question for intelligence is dissonance, which localizes a defect without naming its repair; heuristic anticipation is the agent's search over coherent groupings of its own record, which gropes toward an answer without supplying its content; insight's formal footprint is the refinement (whose direction the analyst models through restoration, minimality, and the complexity ordering); formulation is the mechanism assignment on the new cells; reflective judgment is the agent's consistency test—the faculty distinct from the posterior that asks whether any expressible theory is adequate ([remark 4.2](#)), as against the analyst's warrant accounting, which the agent generally cannot perform on itself ([corollary 8.3](#)); and decision is the standing policy, which then determines what experience will teach next. The election adds the economic layer: an unrestricted desire to know, prosecuted under restricted resources, and eventually surrendered when hope of an answer fails. Nothing in the mathematics represents the conscious act of insight itself; the model represents its products and its price, which is what an economic theory can responsibly claim.

10.3 Limitations and the research program

Four boundaries of the present model define the program's next steps. First, the mechanism vocabulary Θ is fixed: agents lack distinctions, not concepts of consequence-generation. Expanding the vocabulary itself—deeper unawareness—is the natural second paper in the sequence, and the awareness-transition machinery here is built to receive it. Second, the model prices participation in inquiry but not its intensity or duration, and the technology's success is exogenous; endogenizing the depth of inquiry connects to models of costly deliberation and planning (Ryall, 2026), where cached prior conclusions govern when reconsideration is rational—the theory-to-plan transition. Third, agents are individuals; the theory-bearing *firm*—how insights become shared languages, collective intentions, and organizational commitments—is a separate paper resting on the same primitives (Ryall and Wilkins, 2018). Fourth, the competitive consequences here run through payoffs in a fixed game; joining the front end developed here to the cooperative back end of value capture (Gans and Ryall, 2017; Bryan et al., 2022), where awareness itself earns rents and disclosure is strategic, completes the arc. Within the present model, the flagged analytical items are general conditions for the convergence of play within Bayesian worlds (remark 9.7), equilibrium in the anti-coordination region that value destruction creates, counterfactual transparency as a design variable, and Markov state processes.

11 Conclusion

This paper models the step the theory-based view has described but not formalized: the origin of a strategic theory. The model draws a formal boundary between learning within a causal language and producing a new one, and it prices the crossing. Bayesian updating cannot expand the language and cannot even detect that expansion is needed; detection is statistical dissonance, a faculty distinct from the posterior. Dissonance directs repair without determining it; observational equivalence makes any economical inquiry technology fallible; and the agent estimates the value of a question—from its own record, honestly, without the analyst's list of answers—at exactly the moment the acting belief is undefined. The return to Bayesian life is disciplined by a warrant account whose central theorem is easily stated: the insight is not evidence. And because theories cause the actions that cause the evidence, inquiry is almost surely absorbed: eventually every agent either lives in a Bayesian world or rests in a dissonance it has no priced reason to fix—in configurations that preserve false theories, heterogeneous languages, and permanent performance differences. Shared false theories survive exactly when neither strategic variety nor environmental harshness disturbs them; performance gaps persist whenever the laggard's question is priced below the cost of asking. Strategy's oldest question—why do similar firms perform differently?—here receives an answer that begins before beliefs: because they came to different questions, and some questions died, or were abandoned, before they were answered.

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A The evidence process

Fix agent i and theory m . At a visited pair (C, a) write $p = p_i^*(\cdot \mid C, a)$ and $q = \text{marg}_i m(C)(a)$; adequacy at (C, a) is $\text{TV}(p, q) \leq \eta$, equivalently $|p(D) - q(D)| \leq \eta$ for every $D \subseteq X_i$. For each such D and each betting fraction λ , retain the one-sided factor

$$F_{\lambda,D}(x) = 1 + \lambda(\mathbf{1}\{x \in D\} - (q(D) + \eta)).$$

Fix $0 < \lambda_{\max} \leq 1/(1 + \eta)$, choose a countable dense set $\Lambda \subset (0, \lambda_{\max})$, and assign every $\lambda \in \Lambda$ a strictly positive atomic weight, with the weights summing to one. Assign every $D \subseteq X_i$ strictly positive weight as well, and let $M_i^{m,(C,a)}$ be the resulting convex mixture of the fixed processes that multiply by $F_{\lambda,D}(x)$ at each visit to (C, a) and by 1 otherwise. Because D ranges over all subsets of X_i , a negative discrepancy on D is a positive discrepancy on D^c , so no separate sign mixture is needed. Under adequacy, $p(D) \leq q(D) + \eta$ and hence $\mathbb{E}_p[F_{\lambda,D}] \leq 1$, so every fixed component is a nonnegative supermartingale and their convex mixture remains one. Assign every visited-pair component a strictly positive weight and set

$$E_i^m = \sum_{(C,a)} v_{(C,a)} M_i^{m,(C,a)}, \quad \sum_{(C,a)} v_{(C,a)} = 1, \quad v_{(C,a)} > 0.$$

As a convex combination of nonnegative supermartingales, E_i^m is a nonnegative supermartingale under the null that m is adequate at every pair, with $E_i^m = 1$ before play; Ville's inequality then

gives $\Pr(\sup_t E_i^m \geq e^{b_i}) \leq e^{-b_i}$, the false-alarm bound of [definition 5.1](#). If instead m is strictly inadequate at a recurrent pair $j^* = (C, a)$, some D satisfies $p(D) > q(D) + \eta$. For that event define

$$g(\lambda) = \mathbb{E}_p \log(1 + \lambda[\mathbf{1}_D - (q(D) + \eta)]).$$

Then $g(0) = 0$ and $g'(0) = p(D) - q(D) - \eta > 0$, so every sufficiently small positive λ has $g(\lambda) > 0$. Density of Λ supplies a positive-weight atom λ^* with strictly positive expected log-growth; if W^* is its fixed component and $\pi^* > 0$ its weight within the pair mixture, the strong law under recurrent visitation gives $W^*(t) \rightarrow \infty$ almost surely. Because the global process assigns the pair fixed weight $v_{j^*} > 0$, pathwise

$$E_i^m(t) \geq v_{j^*} \pi^* W^*(t) \longrightarrow \infty \quad \text{a.s.}$$

The positive per-period log-growth rate of such a component is its *betting exponent* (and the supremum across available components is the optimal log-growth exponent); this is the dominating-component property used in [lemma 5.2](#) and [proposition 9.10](#).

The exact deterministic case $\eta = 0$ uses a separate zero-likelihood detector rather than the stochastic mixture above. Conceptually, $E_i^m(t) = 1$ until the first observed consequence to which theory m assigns probability zero, at which point $E_i^m(t) = +\infty$ and remains there. A 0-adequate deterministic theory never encounters such an event and is never falsely rejected, whereas a 0-inadequate theory at a recurrent pair encounters a diagnostic impossible consequence almost surely and is rejected at its first occurrence. Thus b_i has no effect in the deterministic case.

B Proofs of Theorem 9.5 and Proposition 9.6

Probability framework. All randomness is carried by three mutually independent i.i.d. families on a common probability space: state draws $\{s_t\}$ over periods $t = 1, 2, \dots$, with law ρ ; consequence randomizers $\{U_t\}$, uniform on $[0, 1]$, from which x_t is generated through a fixed measurable representation of the kernel $\tau(s_t)(a_t)$; and production randomizers $\{V_{i,t}\}$, uniform on $[0, 1]$, drawn when agent i elects inquiry ([assumption 9.4\(ii\)](#)). Let $\mathcal{F}_t$ denote the σ -algebra generated by all randomness through period t . Each agent's standing policy and election indicator in period t are $\mathcal{F}_{t-1}$ -measurable, since they are functions of records, beliefs, and conjectures formed from earlier periods; dissonance and the election in period t are thus decided on the record through period $t - 1$. Conditional on $\mathcal{F}_{t-1}$ and on an election by agent i at a repairable record, production succeeds with probability at least $\underline{\pi}$; at an irreparable record it fails surely.

Part (a). Each successful production strictly refines one agent's partition, and, by cumulative awareness ([assumption 7.1](#)), no partition ever coarsens. The total cell count $\sum_i |\mathcal{P}_i|$ is integer valued, starts at no less than n , never exceeds $n|S|$, and strictly increases at every success.

Hence at most $n(|S| - 1)$ successes occur on every path, and each agent's partition is eventually constant.

Part (b). Fix agent i and let $\Delta\pi_i = \max \pi_i - \min \pi_i$. By [proposition 7.4](#), $\widehat{B}_i \leq T\Delta\pi_i$. If $\Delta\pi_i = 0$, then $\widehat{B}_i = 0$ and no election occurs because $\gamma_i > 0$. Hence suppose $\Delta\pi_i > 0$. The election condition $\lambda_i^{(k)} \widehat{B}_i \geq \gamma_i$ fails whenever $\lambda_i^{(k)} < \gamma_i / (T\Delta\pi_i)$; since $\lambda_i^{(k)}$ is nonincreasing with $\lambda_i^{(k)} \rightarrow 0$ and $\gamma_i > 0$, there is a finite k_i^* beyond which no election occurs at failure count $k \geq k_i^*$. The failure counter resets only on a successful production, and by part (a) there are at most $n(|S| - 1)$ successes across all agents. Between consecutive successes the counter climbs monotonically, so each agent elects at most k_i^* times per success-interval and hence finitely often in total; summing over agents, only finitely many elections occur on every path. ([Assumption 9.4\(ii\)](#) is not needed for election finiteness or for parts (c)–(d); it ensures that an elected repairable question has a positive conditional chance of resolution before abandonment.)

Part (c). Let G^* be the larger of the last refinement period and the last election period, almost surely finite by (a) and (b). After G^* , no language changes and no inquiry cost is paid. Because the election rule is mechanical—agent i elects exactly when it is dissonant and $\lambda_i^{(k_i)} \widehat{B}_i \geq \gamma_i$ —every dissonance event after G^* must fail the participation threshold. All other epistemic activity after G^* is judged-Bayesian updating within the fixed terminal language.

Part (d). The event $E = \{\text{standing profile eventually constant}\}$ is future-dependent, so we handle the conditioning by coupling rather than by conditioning on E directly. Decompose $E = \bigcup_{G_2, \sigma^*} E_{G_2, \sigma^*}$ over the countable set of (period, deterministic profile) pairs, where E_{G_2, σ^*} is the event that the standing profile equals σ^* from period G_2 on and $G_2 > G^*$. Fix (G_2, σ^*) and an agent i . Define an *auxiliary* process that plays σ^* in every period from G_2 on, driven by the same i.i.d. draws (s_t, U_t) ; its post- G_2 draws are independent of $\mathcal{F}_{G_2}$, so almost-sure statements about it hold conditional on the (arbitrary) $\mathcal{F}_{G_2}$ -measurable prefix and carry no conditioning bias. On E_{G_2, σ^*} the actual and auxiliary observations coincide from G_2 on, so any almost-sure property of the auxiliary process holds almost surely on E_{G_2, σ^*} .

For the auxiliary process, let $a_{\sigma^*}(s)$ be the action profile induced by σ^* at state s , and define $S_{\sigma^*}(C, a) = \{s \in C : a_{\sigma^*}(s) = a\}$. Call a pair (C, a) *recurrent* if $S_{\sigma^*}(C, a) \neq \emptyset$ and *frozen* otherwise. A frozen pair is visited only in the finite pre- G_2 prefix, so its count and constraint are constant. A recurrent pair is visited each period with fixed probability $\rho(S_{\sigma^*}(C, a)) > 0$ independently, so its count diverges and, by the strong law on the finite space X_i , its cumulative empirical conditional converges to $p_i^*(\cdot \mid C, a)$; the finite prefix contributes a vanishing fraction. By [assumption 9.4\(i\)](#), [lemma 5.2](#) applies to the auxiliary process: every theory active at G_2 and strictly inadequate under σ^* at a recurrent pair is eventually rejected, every theory inactive at G_2 remains absent, and the active set eventually becomes a survivor subset A_i^* whose retained

members are η -adequate under σ^* at every recurrent pair. Recurrently adequate theories may nevertheless be absent from A_i^* because they were excluded when the current language was initialized, were rejected before G_2 , or are stochastically false-rejected afterward. If $A_i^* \neq \emptyset$, agent i is eventually never dissonant. If $A_i^* = \emptyset$, it is eventually always dissonant; since no election occurs after G^* , the mechanical rule gives $\lambda_i^{(k_i)} \widehat{B}_i < \gamma_i$ at every such period, so agent i is in dissonant stasis (definition 9.2)—including the zero-leverage, abandoned-question, and irreparable cases. Taking the union over the countably many (G_2, σ^*) gives the claim on all of E . $\square$

Proof of Proposition 9.6(i). Suppose awareness and standing policies are eventually constant almost surely, conjectures and reference posteriors converge, and each non-stasis agent's standing policy is a strict no-switch point at the limit (strict slack below κ_i , or the strict U_i -argmax when $\kappa_i = 0$), with the stasis agents' carried-forward policies fixed. Condition (i) of definition 9.1 holds at the limit: by continuity of the finite-dimensional payoff comparison in (μ_i, q_i) , the strict slack persists near the limit—or, when $\kappa_i = 0$, the strict argmax keeps $\sigma_i^* = \sigma_i$ so the gain stays zero—and no switch is triggered, so the eventually constant standing policy is stationary under the switching rule. For condition (ii), empirical conjectures converge to the frequencies induced by the eventually constant profile, giving limit agreement. For condition (iii), a non-stasis agent is eventually never dissonant by theorem 9.5(d), and judged updating keeps the acting belief supported on the eventual active set A_i^* , whose members are η -adequate at every recurrent pair, so the mass on any η -inadequate theory is eventually zero. For condition (iv), elections cease by theorem 9.5(b). The limiting configuration thus satisfies the SCUE conditions for every non-stasis agent with the stasis agents' play held fixed—a partial SCUE. $\square$

Proof of Proposition 9.6(ii-a). With deterministic mechanisms, rejection is irreversible. Because the entry configuration satisfies definition 9.1(iii), its active set contains at least one 0-adequate theory: otherwise every currently active theory would eventually be rejected and, since refill is impossible, the active set would remain empty thereafter. A 0-adequate theory matches every on-path observation and is therefore never rejected. The active set consequently remains nonempty, although it may shrink as initially active inadequate theories are rejected.

Dissonance never occurs; $\widehat{B}_i = 0$ at every generated record by hypothesis, so no election or refinement occurs; and strict optimality under every conjecture and every theory in the entry active set fixes σ_i as that set shrinks and as reference posteriors and empirical conjectures evolve. Empirical conjectures converge to induced play, and any initially active inadequate theory is eventually rejected. The SCUE conditions therefore continue to hold. $\square$

Proof of Proposition 9.6(ii-b). Condition on the entry history $\mathcal{F}_{t_0}$ and couple the actual process to the auxiliary process that holds the configuration's standing profile fixed after t_0 ; on the

persistence event below the two coincide. Fix an agent i and an entry-active theory m : by hypothesis it is η -adequate under the fixed profile and has $E_i^m \leq 1$ at entry. Adequacy makes $(E_i^m)_{t \geq t_0}$ a nonnegative supermartingale, so Ville’s inequality gives

$$\Pr\left(\sup_{t \geq t_0} E_i^m \geq e^{b_i} \mid \mathcal{F}_{t_0}\right) \leq E_i^m(t_0) e^{-b_i} \leq e^{-b_i},$$

so m is never rejected off an event of probability at most e^{-b_i} . Agent i becomes dissonant only if *every* entry-active theory is rejected—an event contained in “ m is rejected” for any single m —so that too has probability at most e^{-b_i} ; a union bound over agents leaves probability at least $1 - \sum_i e^{-b_i}$ that no agent is ever dissonant. On that event each active set stays nonempty and, by permanence of rejection, remains contained in the entry set and hence supported on η -adequate theories. The strict-optimality hypothesis then fixes each standing action under every conjecture and every acting belief judged updating can produce, so $\sigma_i^* = \sigma_i$ and no switch fires; a nonempty active set makes $\hat{B}_i = 0$, so no election fires. Play therefore never changes, empirical conjectures converge to the induced play, and the SCUE conditions hold in every period. In the deterministic case $\eta = 0$ every false-alarm probability is zero and the conclusion holds surely. $\square$

C Supporting computations for the running example

Dominance thresholds. At an awareness cell C , write $p^a = \mu_i(\{m : m(C) = \theta^a\})$ for $a \in \{0, 1\}$. Against a rival playing 0, the subjective expected-payoff difference between own actions 0 and 1 is $\frac{1}{2}p^0 - (1 - \frac{\beta}{2})p^1$; against a rival playing 1 it is $(1 - \frac{\beta}{2})p^0 - \frac{1}{2}p^1$; mixtures follow by linearity. Action 0 is strictly dominant on C if and only if $p^0 > (2 - \beta)p^1$, and symmetrically. Certainty in an architecture label makes the labeled action strictly dominant for every $\beta \in [0, 1]$, which is what sustains the standing policies in the trap configuration. At $\beta = 1$ the threshold is simple belief asymmetry, which is what differentiates the firms’ opening policies.

Per-token gain and deliberative leverage. Consider a retained token whose coherent group pins mechanism θ^{a^*} while the standing policy plays $a \neq a^*$ at the token’s cell, at $\beta = 1$. Proposing a^* : if the rival plays a^* , the standing action yields 0 (differentiated loss) and a^* yields $\frac{1}{2}$ (pool); if the rival plays a , the standing action yields $\frac{1}{2}$ (pool) and a^* yields 1 (differentiated win). The per-token robust gain is $\frac{1}{2}$ in both cases—transparency makes $\Theta_i(g)$ a singleton, so the mechanism minimum is degenerate. A token whose group’s optimal action equals the standing one contributes 0. At discovery, the winning token contributes zero and the losing token contributes $\frac{1}{2}$ to the fine value, while under the coarse one-cell current language a uniform

reversal gains $\frac{1}{2}$ on the losing token and loses $\frac{1}{2}$ on the winning token. Thus

$$L^{\text{fine}} = \frac{1}{2}, \quad L^{\text{cur}} = 0, \quad J = \frac{1}{2}, \quad \widehat{B} = (2/2)(\frac{1}{2}) = \frac{1}{2}.$$

After the validation stretch of [section 7](#) (serving the two off-diagonal states, valuation horizon $T = 2$), the later winner has

$$L_1^{\text{fine}} = L_1^{\text{cur}} = J_1 = \widehat{B}_1 = 0.$$

The later loser's four tokens split into two pinned-mechanism groups; two disagree with its column policy and contribute $\frac{1}{2}$ each under fine contingency, while two agree and contribute zero. Within either current column a uniform reversal gains $\frac{1}{2}$ on one token and loses $\frac{1}{2}$ on the other, so

$$L_2^{\text{fine}} = 1, \quad L_2^{\text{cur}} = 0, \quad J_2 = 1, \quad \widehat{B}_2 = (2/4)1 = \frac{1}{2}.$$

The durable content is the asymmetry $\widehat{B}_1 = 0 < \widehat{B}_2$, not the coincidence that $\widehat{B}_2$ equals the discovery value.

The trap and its knife edge. With both firms certain of the column assignment, the column policy pools at every state; every payoff is $\frac{1}{2}$; and under pooled play every mechanism consistent with the column cells predicts $\frac{1}{2}$ at $\beta = 1$. No token conflicts, active sets never empty, and $\widehat{B}_i = 0$; conditions (i)–(iv) of [definition 9.1](#) hold. For $\beta < 1$, at an off-diagonal state the pooled action is objectively wrong and the realized payoff is $\beta/2$ against a prediction of $\frac{1}{2}$; the payoff $\beta/2$ uniquely reveals that the other architecture is preferred, while a payoff of $\frac{1}{2}$ would leave preferred and neutral both possible. The revealed conflict makes one column cell incoherent, the active set empties, and the reflective posterior over the row and column worlds becomes degenerate: dissonance, leverage, and warrant closure arrive together.

Persistence payoffs. Take $\beta < 1$, firm 1 playing the true row assignment, firm 2 coarse and playing a constant architecture c . The firms differentiate at the two states of the row that c misrepresents—firm 1 wins both—and pool at the other row's two states, giving firm 1 average payoff $\frac{3}{4}$ and firm 2 $\frac{1}{4}$ under uniform ρ . Firm 2's cumulative record rejects all three constant theories: its differentiated losses defeat the constant theory matching c , its pooled payoff of one-half defeats the opposite theory's $\beta/2$ prediction, and neutrality fails at the differentiated losses, so it is genuinely dissonant. For the balanced four-token record, the two correctly served tokens contribute zero and the two mis-served tokens contribute $\frac{1}{2}$ each, so $L_2^{\text{fine}} = 1$. A uniform current-language reversal yields $\beta - 1 < 0$, while retaining the standing action yields zero, so

$$L_2^{\text{cur}} = 0, \quad J_2 = 1, \quad \widehat{B}_2 = (2/4)1 = \frac{1}{2}.$$

For other realized compositions its deliberative leverage is positive when the representational increment is positive and remains bounded, and stasis obtains whenever $\gamma_2 > \lambda_2^{(k_2)} \hat{B}_2$. (At $\beta = 1$ the same constant- c record leaves the opposite constant theory adequate, which prescribes switching c ; firm 2 then oscillates rather than resting, so the persistent stable difference is specific to the diagnostic environment $\beta < 1$.)